

v7 cloning campaign: pre-training, in-sample optimization and held-out generalization

2026-09-08

State of the results on 2026-09-08. Every number in this report is read from a file of the pulled runs (`~/Documents/Research/xcquinox-results/runs/dfs_step7/`) or from a CSV the figure suite wrote beside the figure it belongs to; the file is named at each table. The figures are the tracked sets under `notebooks/analysis/figures_dfs_step7_*`; the validation-best checkpoint channel is used for every optimization figure unless a caption says otherwise.

0 Scope and inventory

The v7 campaign trains one network pair (an exchange network and a correlation network) per architecture and per training-subset size, in three groups that share one protocol and differ in their architecture axis. Each group runs its own pre-training (the cloning of the parent functional), a fidelity certificate that gates the group’s training array, and the training array itself, one cell per (architecture, subset size).

Group	Configuration file	Architectures	Cells done / planned	Run
G1 (size)	<code>dfs_step7.dfs6311_grid3_v7g1_size.yaml</code>	shallow, shallow_attn, medium, medium_attn	medium 11/11, medium_attn 7/11, shallow 0/11, shallow_attn 0/11	run_20260902T145245Z
G2a (descriptor families, GGA core)	<code>dfs_step7.dfs6311_grid3_v7g2a_families_core.yaml</code>	deep_3x16, deep_attn_3x16, deep_cusp_3x16	deep_3x16 10/11, deep_attn_3x16 0/11, deep_cusp_3x16 0/11	run_20260902T145247Z
G2 (descriptor families, meta-GGA)	<code>dfs_step7.dfs6311_grid3_v7g2_families_mgga.yaml</code>	deep_cusp_mgga_3x16, deep_mgga_3x16, deep_mgga_attn_3x16, deep_rung35_mgga_3x16, deep_rung35ms_mgga_3x16	0/55 (the first certificate reads FAIL, Sec. 2.4; the other four are pending)	run_20260902T145250Z

The subset-size axis is 1, 2, 3, 4, 5, 6, 7, 12, 15, 18 and 26 training molecules, drawn from the 26-point DFS training pool by the Jensen-Shannon subset selection; the subset memberships are printed in the footer of the held-out energy figure (Fig. 4.1). Basis 6-311++G(3df,2pd), grid level 3, density fitting on, for every stage (`fidelity_certificate.json`, keys `basis`, `grid_level`, `density_fit`). The cells that have not finished are the attention and cusp architectures at the larger subsets and the whole meta-GGA group; they appear as “pending” in the tables below and are absent from the figures. Two arms of a follow-up (the 25-cycle SCF arm and the dpyscf-parity arm, both on the medium architecture) were submitted on 2026-09-08 and carry no results yet.

1 Model, descriptors and the functional form

Sources: `xcquinox/alec/networks.py` (the two networks), `xcquinox/alec/descriptors.py` and `xcquinox/alec/metagga.py` (the extra descriptors), `xcquinox/net.py` (the attention block), `xcquinox/alec/config`.

py (the architecture registry). The coordinate convention is the DFS one (model.descriptor_coordinates:dfs in every v7 file), following Dick and Fernandez-Serra, PRB 104, L161109 (2021), whose equation numbers are quoted as "DFS Eq. n".

1.1 Density variables

With $n = n_\alpha + n_\beta$ the total density and $\sigma = |\nabla n|^2$,

$$k_F = (3\pi^2 n)^{1/3}, \quad s = \frac{|\nabla n|}{2k_F n}, \quad r_s = \left(\frac{3}{4\pi n}\right)^{1/3}, \quad \zeta = \frac{n_\alpha - n_\beta}{n}, \quad \phi(\zeta) = \frac{1}{2} \left[(1 + \zeta)^{4/3} + (1 - \zeta)^{4/3} \right].$$

s is the reduced density gradient (DFS Eq. 5), ϕ the spin-scaling factor of DFS Eq. 4 ($\phi = 1$ at $\zeta = 0$, $2^{1/3}$ at full polarization). The density is floored at the network's lower_rho_cutoff before any of these is formed.

1.2 Network coordinates (DFS coordinates)

The exchange network reads one coordinate at the GGA level and two at the meta-GGA level; the correlation network reads three and four. They are logarithmic compressions of the variables above (DFS Eqs. 7 to 10):

$$x_0 = \ln(n^{1/3} + 10^{-5}), \quad x_1 = \ln \phi(\zeta), \quad x_s = (1 - e^{-s^2}) \ln(s + 1), \quad x_\alpha = \ln \frac{\alpha + 1}{2}.$$

Exchange input: $[x_s, \dots]$, where s is that of the doubled spin channel $2n_\sigma$ (the exchange network is evaluated per spin channel, Sec. 1.4). Correlation input: $[x_0, x_1, x_s, \dots]$ on the total density. The ellipsis holds the extra descriptors of the architecture, in the registry's list order; on the meta-GGA rung the raw α column among them is replaced by x_α .

1.3 Extra descriptors

Iso-orbital indicator (meta-GGA rung, metagga), SCAN's α (Sun, Ruzsinszky and Perdew, PRL 115, 036402 (2015), Eq. 2; DFS Eq. 6):

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \quad \tau = \frac{1}{2} \sum_i |\nabla \psi_i|^2 = \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} \nabla \chi_\mu \cdot \nabla \chi_\nu, \quad \tau_W = \frac{|\nabla n|^2}{8n}, \quad \tau_{\text{unif}} = \frac{3}{10} (3\pi^2)^{2/3} n^{5/3}.$$

τ is a contraction of the live one-particle density matrix P against the AO gradients, so it is self-consistent and differentiable through the SCF without second derivatives. The bound $\alpha \geq 0$ (the von Weizsacker inequality) is imposed by a smooth positive part rather than a hard clip, so the descriptor and its derivative stay continuous where a spin channel holds one orbital.

Nuclear cusp pair (cusp), two bounded features per grid point:

$$x_4 = \exp(-2Z_{\text{nearest}} r_{\text{min}}), \quad x_5 = \tanh\left(\frac{1}{5} \ln \sum_A \frac{Z_A}{r_A}\right).$$

x_4 is a proximity feature following the density-form cusp envelope e^{-2Zr} (Steiner, J. Chem. Phys. 39, 2365 (1963), the density counterpart of Kato's wavefunction cusp); x_5 is the log-compressed bare-nuclear potential $-V_{\text{ext}}$, the log and the 1/5 scale being the conditioning of this work. Both are geometric, so the synthetic pre-training mesh (Sec. 2.1) carries no values for them.

The remaining registry descriptors (density-matrix statistics, the rung-3.5 occupancies) are not on any v7 axis and are not described here.

1.4 The functional form

Each network is a multilayer perceptron with GELU activations, depth hidden layers of nodes units, a final linear layer (initialized at zero in the `deep_*` family, `zero_init_final_layer`), and an output squash. With g the MLP output at a grid point,

$$F_x = 1 + \mathcal{L}_{1.804}(G_x \cdot g_x), \quad F_c = 1 + \mathcal{L}_2(G_c \cdot g_c), \quad \mathcal{L}_\lambda(z) = \lambda \operatorname{sigmoid}(z - \ln(\lambda - 1)) - 1.$$

$\mathcal{L}_\lambda(0) = 0$, so a zero MLP output returns $F = 1$ (the local density approximation), and F stays in $(0, \lambda)$ for every z . $\mathcal{L}_\lambda$ is the DFS I_a map (DFS Eq. 11) to machine precision. For exchange $\lambda = 1.804 = 1 + \kappa_{\text{PBE}}$, the PBE Lieb-Oxford-motivated ceiling (PBE, PRL 77, 3865 (1996), Eq. 14); DFS uses 1.174 (the tighter bound of Perdew, Ruzsinszky, Sun and Burke, J. Chem. Phys. 140, 18A533 (2014)), and the v7 meta-GGA architectures keep 1.804 as well (`config.resolved_xnet_lob_lim`), a recorded deviation. For correlation $\lambda = 2$ is the symmetric sigmoid of DFS Eq. 13; its purpose is $F_c \geq 0$, the upper value 2 is incidental.

G is the uniform-electron-gas gate that pins $F = 1$ where the gradient vanishes:

$$G = \tanh^2(s) \text{ (GGA rung)}, \quad G = x_s + \tanh^2(x_\alpha) \text{ (meta-GGA rung, DFS Eq. 12)},$$

evaluated on the raw s (and α), never on the compressed coordinate. On the meta-GGA rung the same gate form is used for both networks (DFS’s correlation gate applies $\tanh$ to x_s ; recorded deviation).

The exchange-correlation energy is assembled with the exchange spin-scaling relation of Oliver and Perdew (PRA 20, 397 (1979)) and the PW92 correlation baseline:

$$E_{xc} = \frac{1}{2} (E_x[2n_\alpha] + E_x[2n_\beta]) + E_c[n_\alpha, n_\beta], \quad E_x[n] = \int n \epsilon_x^{\text{LDA}}(n) F_x d^3r, \quad E_c = \int n \epsilon_c^{\text{PW92}}(r_s, \zeta) F_c d^3r,$$

so the exchange network sees each spin channel at its doubled density (its x_s is that of $2n_\sigma$), and the correlation network sees the total density with ζ through x_1 .

Attention variants. In the `_attn` architectures a self-attention block (Vaswani et al., 2017, Eqs. 1 and 2, pre-layer-norm as in Xiong et al., 2020) sits after the first hidden layer. The hidden vector of H units at a grid point is treated as h tokens of dimension $d_k = H/h$ and attention runs across the token axis,

$$\text{Attn}(Q, K, V) = \operatorname{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad x \leftarrow x + W^O \operatorname{Concat}(\text{head}_1, \dots, \text{head}_h)(\text{LN}(x)).$$

It is a per-point re-mixing of hidden units, not a spatial or non-local operation.

1.5 Architectures on the v7 axes

Architecture	Rung, parent	Hidden layers x units	Attention heads	Extra descriptors	Exchange input	Correlation input
shallow	GGA, PBE	2 x 8	none	none	x_s	x_0, x_1, x_s
shallow_attn	GGA, PBE	2 x 8	2	none	x_s	x_0, x_1, x_s
medium	GGA, PBE	3 x 16	none	none	x_s	x_0, x_1, x_s
medium_attn	GGA, PBE	3 x 16	4	none	x_s	x_0, x_1, x_s
deep_3x16	GGA, PBE	3 x 16	none	none	x_s	x_0, x_1, x_s
deep_attn_3x16	GGA, PBE	3 x 16	4	none	x_s	x_0, x_1, x_s
deep_cusp_3x16	GGA, PBE	3 x 16	none	cusp	x_s, x_4, x_5	x_0, x_1, x_s, x_4, x_5
deep_cusp_mgga_3x16	meta-GGA, SCAN	3 x 16	none	cusp, α	x_s, x_4, x_5, x_α	$x_0, x_1, x_s, x_4, x_5, x_\alpha$

medium and deep_3x16 share the layer geometry; the deep_* family carries the physics-correction flags of the registry (dm_entropy_intensive, descriptor_log_transform, zero_init_final_layer), of which only the zero initialization and the cusp column-1 logarithm act on this axis (the DFS coordinates already compress s). All v7 networks are unanchored (model.parent_anchor:false): the network output is the whole enhancement factor, and the parent functional enters only as the pre-training target.

2 Pre-training (functional cloning)

Sources: xcquinox/alec/pretrain.py, xcquinox/alec/pretrain_data_gen.py, the pretrain: and fidelity: blocks of the resolved configurations, pretrain/<arch>/pretrain_metadata.json and pretrain/<arch>/fidelity_certificate.json of each run.

2.1 Set, targets and objective

Each network is fitted to reproduce its parent functional’s enhancement factor pointwise on the parent’s own self-consistent densities: PBE for the GGA rung, SCAN for the meta-GGA rung. The set is the DFS pre-training inventory (8 free atoms and 22 G2/97 molecules at the GGA level; 20 molecules at the meta-GGA level, which drops H2 and N2) plus every single-atom species of the BH76 and W4-11 pools at its production charge and spin (the 12 neutral elements and the F- and Cl- anions), 39 systems for the GGA groups and 38 for the meta-GGA group (n_systems of the certificates). The targets are stored per grid point as

$$t_x = F_x^{\text{parent}} - 1 = \frac{\epsilon_x^{\text{parent}}}{\epsilon_x^{\text{LDA}}} - 1, \quad t_c = F_c^{\text{parent}} - 1 = \frac{\epsilon_c^{\text{parent}}}{\epsilon_c^{\text{PW92}}} - 1,$$

from spin-resolved libxc evaluations, the exchange rows posed per spin channel at the doubled density ($2n_\sigma, 4\sigma_{\sigma\sigma}$), the footing the production exchange evaluates (exchange_footing:spin_channel). A synthetic (r_s, s, α) mesh (the DFS SI’s (s, α) grid extended to three dimensions) is added as a regularizer at a fixed 30 percent share of the integration weight (mesh_fraction:0.3); its rows belong to no system.

The objective per network is the integration-weighted pointwise residual plus a per-system energy term (_PretrainLoss):

$$\mathcal{L}_{\text{pre}} = \frac{\sum_i w_i (F_i^{\text{NN}} - 1 - t_i)^2}{\sum_i w_i} + w_E \frac{1}{N_{\text{sys}}} \sum_s \left(\sum_{i \in s} w_i^{\text{grid}} n_i \epsilon_i^{\text{LDA}} F_i^{\text{NN}} - E_{xc,s}^{\text{parent}} \right)^2, \quad w_i = |n_i \epsilon_i^{\text{LDA}}| w_i^{\text{grid}}, \quad w_E = 0.1.$$

The energy term exists because the pointwise residual alone does not bound a system’s energy: across the earlier campaign the architecture with the lowest exchange residual carried the largest atomization-energy offset from its parent. Schedule: Adam, 20000 steps, learning rate 10^{-3} decaying linearly to 10^{-5} from the midpoint, gradient clip 1.0, a 20 percent held-out validation slice scored every 50 steps with patience 300 checks (pretrain: block). The curves of Fig. 2.1 are the pointwise term per step taken.

2.2 The fidelity certificate

After the fit, the saved networks are re-evaluated on every system of the set at the production basis and grid and compared with the parent: the per-system exchange-correlation energy error $\Delta E_{xc,s} = E_{xc,s}^{\text{NN}} - E_{xc,s}^{\text{parent}}$, and, for every molecule, the atomization-energy error $\Delta AE = \Delta E_{xc,\text{mol}} - \sum_A \Delta E_{xc,A}$ against the parent’s own atomization energy. The gate (fidelity: block, two-tier since 2026-09-02) is

$$\max_{\text{atoms}} |\Delta E_{xc,A}| \leq 1.0 \text{ mHa}, \quad \text{mean}_{\text{mol}} |\Delta AE| \leq 1.0 \text{ kcal/mol}, \quad \max_{\text{mol}} |\Delta AE| \leq 2.0 \text{ kcal/mol},$$

and a failing certificate makes the pre-training task exit without success, which holds the group’s preflight and training array. The certificate is what makes the training-stage comparison against

PBE (or SCAN) meaningful: every trained cell starts from a network that is its parent to within these tolerances.

2.3 Figures

Fig. 2.1 Pre-training loss curves, X-net (exchange F_X) and C-net (correlation F_C), one line per architecture, log-y; the spikes are the validation checks' snapshot reloads.

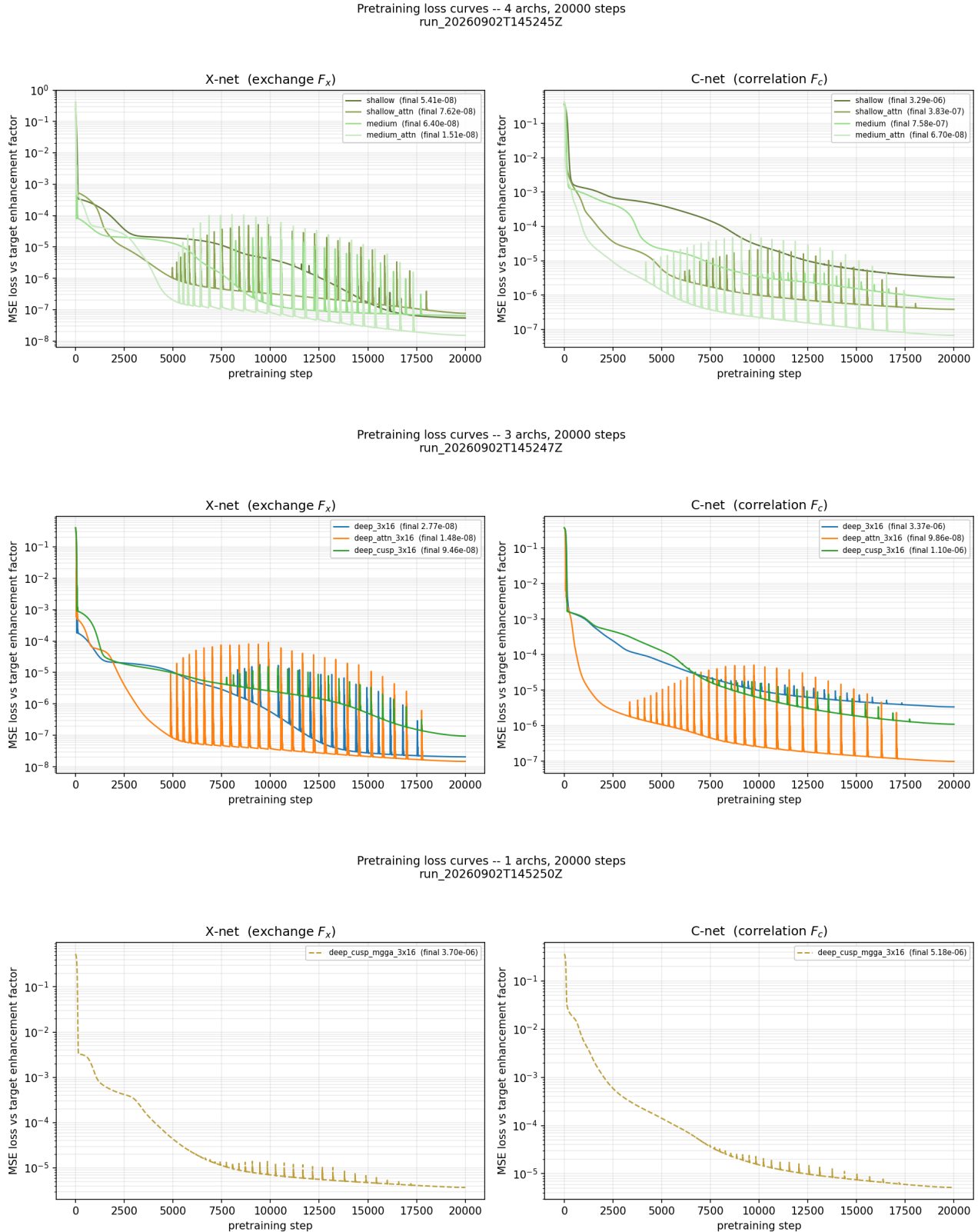


Fig. 2.2 Certificate summary: mean $|\Delta AE|$ per architecture (bar) with the per-species maximum (marker), against the 1.0 kcal/mol mean gate and the 2.0 kcal/mol per-species backstop; the v7 round beside the legacy (pre-protocol, single-tolerance) round of the same G1 architectures. The meta-GGA certificate is tabulated in Sec. 2.4 (the summary figure was rendered before it existed).

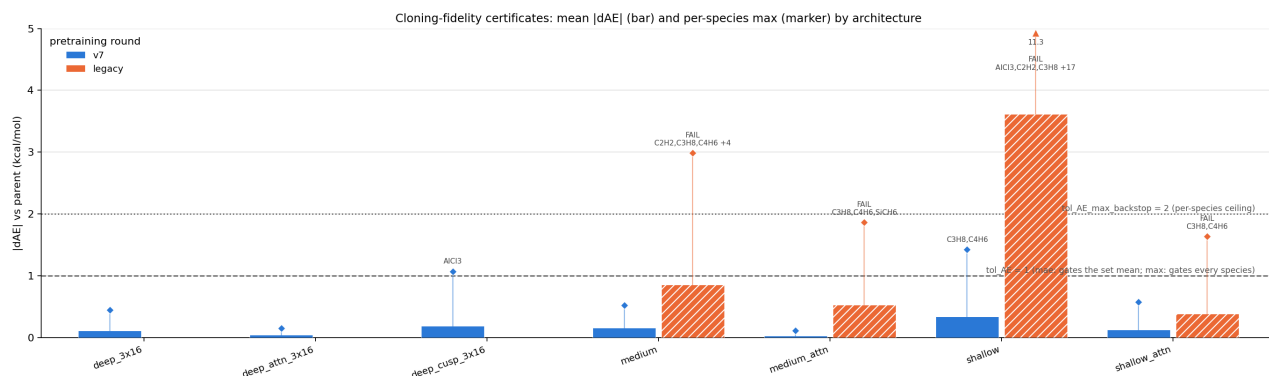
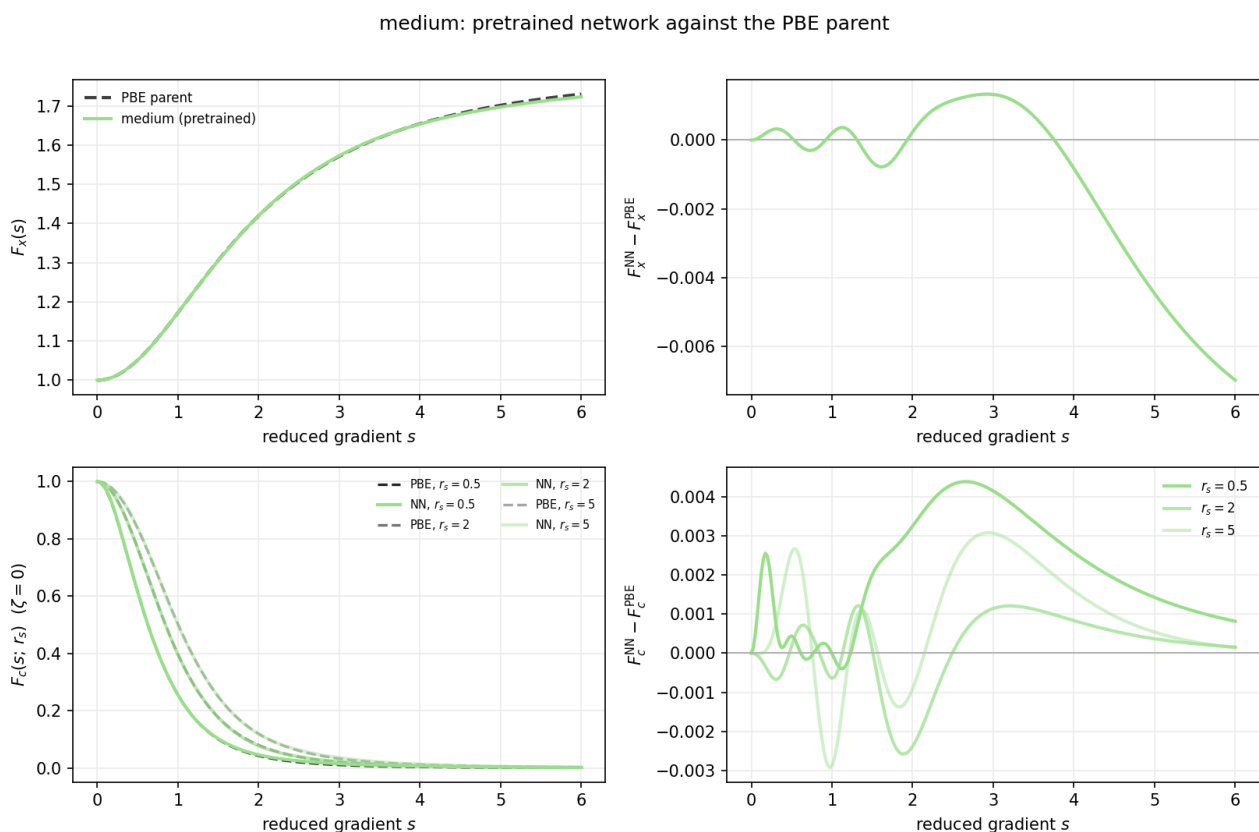
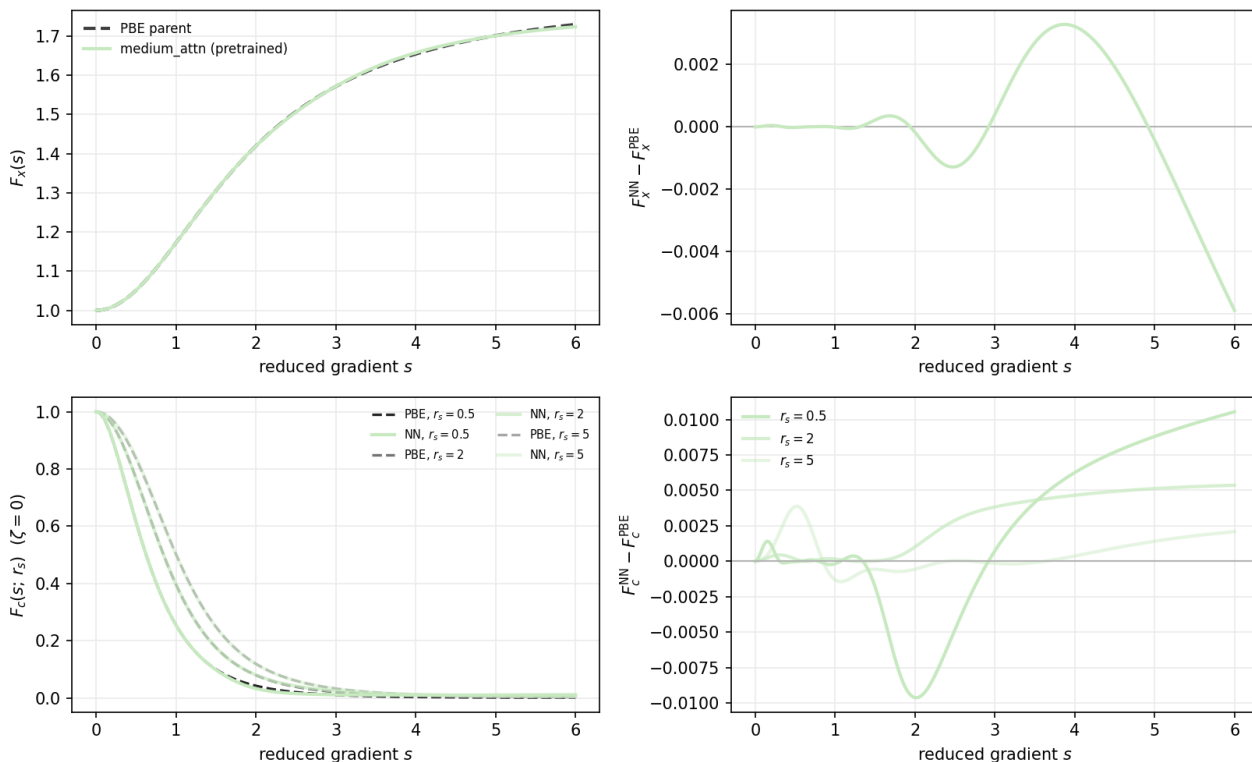


Fig. 2.3 Pre-trained enhancement factors against the parent, per architecture: $F_x(s)$ at $n = 1$ and $F_c(s; r_s)$ at $\zeta = 0$, $r_s \in \{0.5, 2, 5\}$, with the difference panels $F^{\text{NN}} - F^{\text{parent}}$. For the meta-GGA clone the slices are at $\alpha = 0$ and $\alpha = 1$ (the SCAN Fig. 1 convention). The parent curves are the anchor's own libxc-constant implementations.

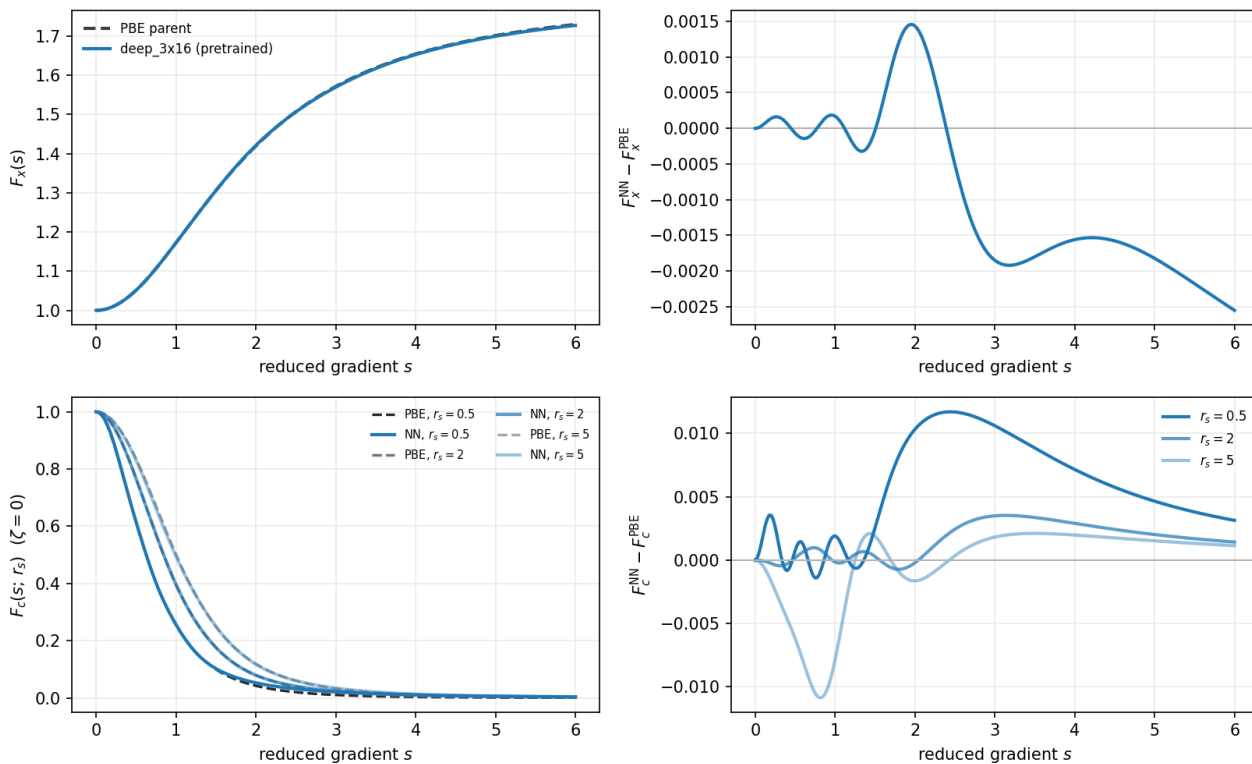


run run_20260902T145245Z; slices: F_x at rho=1, zero extra descriptors; F_c at zeta=0; max|dF_x| 6.98e-03. certificate PASS: max atom 2.10e-01 mHa, max dAE 5.19e-01 kcal/mol (parent pbe)

medium_attn: pretrained network against the PBE parent



deep_3x16: pretrained network against the PBE parent



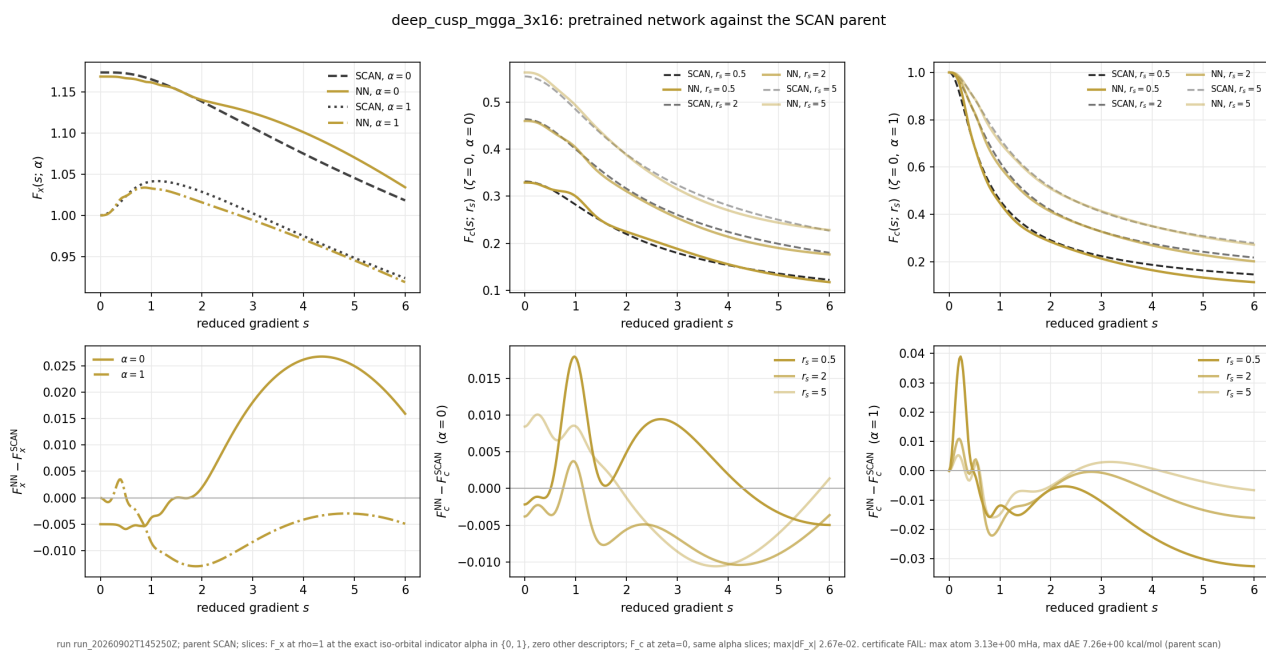
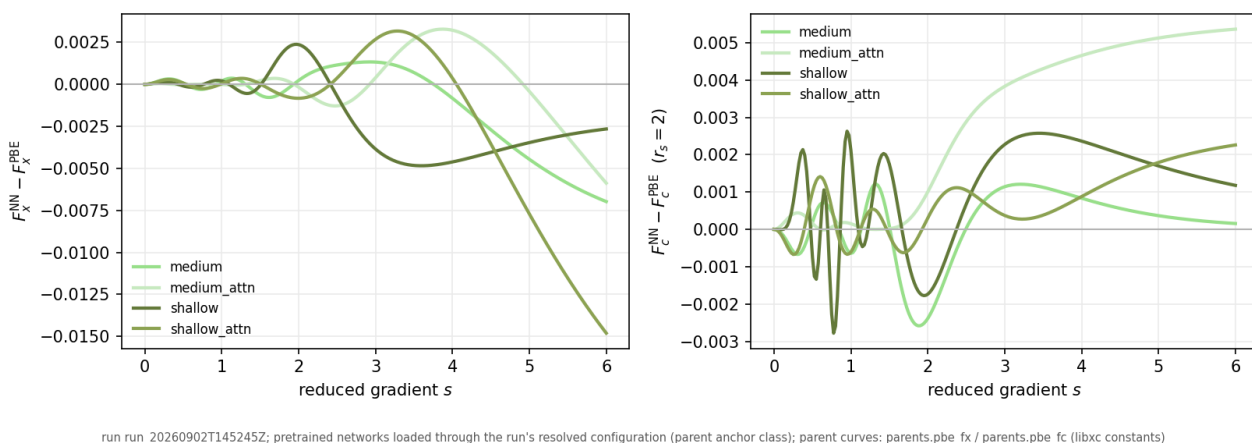
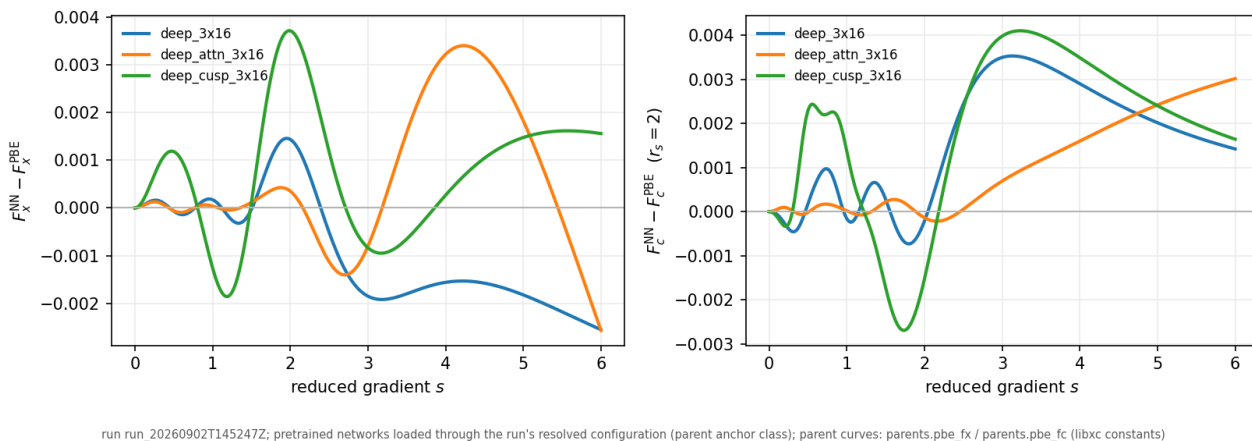


Fig. 2.4 The learned corrections of every GGA architecture on one axis (G1 and G2a).

Pretrained corrections to the PBE parent, all architectures



Pretrained corrections to the PBE parent, all architectures



The shallow pair and the deep_attn_3x16 and deep_cusp_3x16 clones are drawn in the same directories (pretrain_fx_fc_arch.png).

2.4 Certificate results

Read from `pretrain/<arch>/fidelity_certificate.json` and `pretrain_metadata.json`; the pointwise losses are the saved network's, in units of the squared enhancement-factor residual.

Group	Architecture	Parent	Steps	Pointwise loss X / C	max atom $ \Delta E_{x_C} $ (mHa)	mean $ \Delta A E $ (kcal/mol)	max $ \Delta A E $ (kcal/mol)	Species over 1 kcal/mol	Verdict
G1	shallow	PBE	20000	5.41e-08 / 3.29e-06	0.369	0.343	1.421	C3H8, C4H6	PASS
G1	shallow_attn	PBE	20000	7.62e-08 / 3.83e-07	0.201	0.131	0.573	none	PASS
G1	medium	PBE	20000	6.40e-08 / 7.58e-07	0.210	0.162	0.519	none	PASS
G1	medium_attn	PBE	20000	1.51e-08 / 6.70e-08	0.058	0.031	0.115	none	PASS
G2a	deep_3x16	PBE	20000	2.77e-08 / 3.37e-06	0.439	0.114	0.450	none	PASS
G2a	deep_attn_3x16	PBE	20000	1.48e-08 / 9.86e-08	0.123	0.042	0.149	none	PASS
G2a	deep_cusp_3x16	PBE	20000	9.46e-08 / 1.10e-06	0.439	0.188	1.071	AlCl3	PASS
G2	deep_cusp_mgga_3x16	SCAN	20000	3.70e-06 / 5.18e-06	3.132	2.062	7.257	AlCl3, C2H2, C3H8, C4H6, CH2, CH4, CO, CO2, F2, H2O, HCN, O2, PH3, SiCH6	FAIL

Every GGA clone passes on all three criteria; the two with a species above 1 kcal/mol (shallow: C3H8, C4H6; deep_cusp_3x16: AlCl3) pass on the mean and under the 2.0 kcal/mol backstop. The attention variants are the tightest clones by every measure (medium_attn: max atom 0.058 mHa, mean 0.031 kcal/mol; deep_attn_3x16: 0.123 mHa, 0.042 kcal/mol), at the price of the longest fits (about 28 h against 3.6 to 3.8 h for their plain twins, duration_seconds).

The meta-GGA clone deep_cusp_mgga_3x16 fails all three criteria after a converged 20000-step fit (pointwise residuals 3.7e-6 and 5.2e-6, an order of magnitude above the GGA clones’): the largest free-atom error is 3.13 mHa (O; B 2.91, Be 2.61), the mean atomization error 2.06 kcal/mol and the largest 7.26 kcal/mol (C4H6; C3H8 7.19, O2 4.77, SiCH6 3.52, CH4 3.47), 14 of 22 molecules above 1 kcal/mol, every one of the eight largest errors of the same sign. Fig. 2.3 (bottom) shows where the clone misses SCAN: the $\alpha = 0$ exchange slice is 0.015 to 0.027 above SCAN for $s > 2$ and the $\alpha = 1$ correlation slices sit 0.01 to 0.03 below it at $r_s = 0.5$, systematic offsets that do not cancel between a molecule and its atoms. The v6 meta-GGA certificates, all PASS, were fitted in the anchored model class (the network learns a correction on top of SCAN); this is the first unanchored meta-GGA clone the two-tier gate has judged. The four remaining meta-GGA architectures are pending; whether the rung or this architecture is the cause is decided by their certificates. Until then the meta-GGA training array is held.

3 Optimization, in-sample

Sources: xcquinox/alec/losses.py (L5_gradnorm_vxc_step7, the channel terms), xcquinox/alec/train.py (the per-molecule loop, the optimizer), xcquinox/alec/oneshot.py (the SCF tail window), xcquinox/alec/solver_manual.py (the differentiable SCF), the hyperparams: and solvers: blocks of the resolved configurations, and per cell train_metadata.json, losses.npy, aux_log.pkl, eval/per_molecule.json.

3.1 Training protocol

Each cell trains the certified clone of its architecture on a subset of the 26-point DFS pool. The per-molecule update scheme takes one optimizer step per training group (a reaction, or a molecule with its density and potential references) per epoch, so an epoch is one pass over the groups. The loss of a group is the fixed weighted sum

$$\mathcal{L} = 1 \cdot \mathcal{L}_{AE} + 1 \cdot \mathcal{L}_{rxn} + 1 \cdot \mathcal{L}_{IP} + 1 \cdot \mathcal{L}_{v_{xc}} + 20 \cdot \mathcal{L}_{\rho},$$

the Letter’s $\lambda_{RE} = 1$, $\lambda_n = 20$ structure (effective_channel_weights; the GradNorm balancer is configured but dormant under this scheme, and the vxc_weight 0.01 and density_weight 0.1 pre-scales are forced to 1). The channels, with E^{NN} the self-consistent total energy of the 3-cycle SCF (Sec. 3.2) and $t = 0, \dots, N - 1$ the cycle index:

- Reaction energies (ae_as_reactions:true routes the W4-11 atomizations here as molecule-to-atoms reactions with the network’s own atoms, beside the BH76 barriers):

$$\mathcal{L}_{rxn} = \frac{1}{|T|} \sum_{t \in T} w_t^2 r_t^2, \quad r_t = \sum_i c_i E_{i,t}^{NN} - E^{\text{ref}}, \quad w_t = \left(\frac{t}{N-1} \right)^2,$$

T the kept SCF tail (the last $\min(N, 10)$ cycles; at $N = 3$ all three, $w_t^2 = 0, 1/16, 1$), the DFS convergence-tail scoring (DFS Eqs. 15 and 16) that penalizes a non-converging SCF instead of one arbitrary final cycle. Absolute Ha² (loss_metric:absolute). References: W4-11 atomization energies and the W2-F12 BH76 barrier heights of GMTKN55.

- Atom anchors (the residue of the AE channel once the atomizations are reactions): the free atoms of the subset are held near their exact non-relativistic energies E_Z^{anchor} (Chakravorty et al., PRA 47, 3649 (1993)), $\mathcal{L}_{AE} = 0.01 \cdot \text{mean}_Z (E_Z^{NN} - E_Z^{\text{anchor}})^2 / (E_Z^{\text{anchor}})^2$.
- Ionization potentials, $\mathcal{L}_{IP} = \text{mean} (E_+^{NN} - E_0^{NN} - IP^{\text{ref}})^2$, present in the form but empty on every v7 subset (no IP13 point is selected).

- Exchange-correlation potential, on the reference density (frozen features): $\mathcal{L}_{v_{xc}} = \|V_{xc}^{NN} - V_{xc}^{ref}\|_F^2 / n_{AO}^2$, summed over spin channels for open shells.
- Density, the final mixed density of the SCF against the CCSD reference density on the same grid, per electron squared (the Letter's Eq. 17 normalization; `density_per_electron:true`):

$$\mathcal{L}_\rho = \text{mean}_{\text{mol}} \frac{\sum_g w_g (n_g^{NN} - n_g^{ref})^2}{N_e^2}, \quad N_e = \sum_g w_g n_g^{ref}.$$

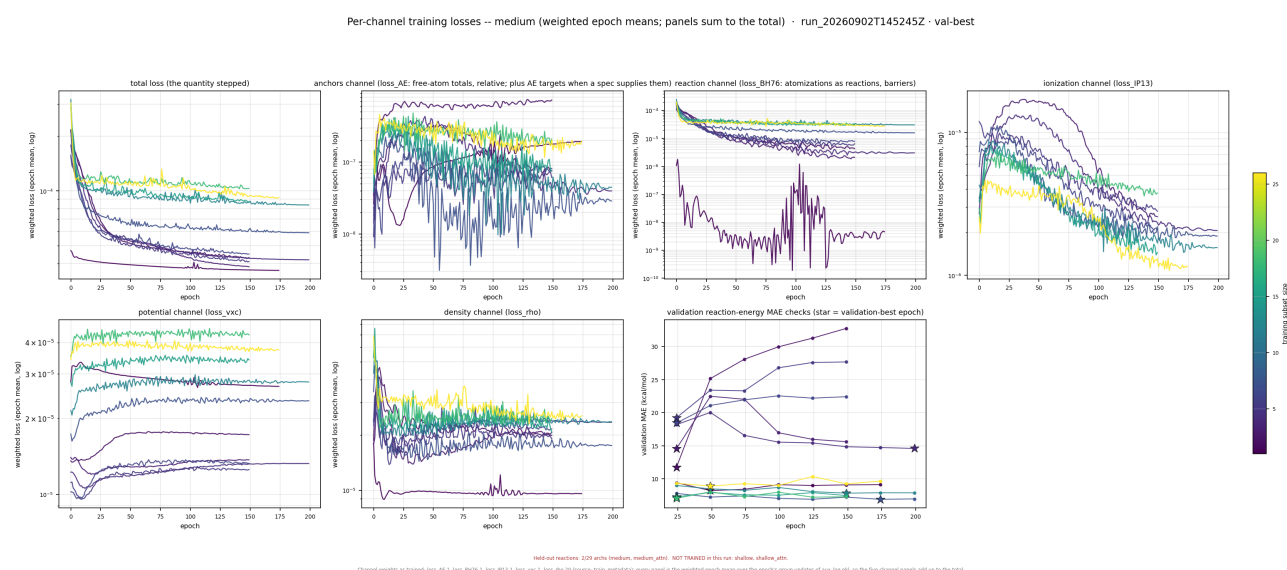
Optimizer: AdamW, learning rate 10^{-3} held for the first half of the run and decayed linearly to 10^{-5} over the second (`lr_decay_start:0.5`), weight decay 10^{-4} , gradient clip 1.0. 200 epochs (`n_steps`), a validation reaction-energy MAE on the 47-reaction validation slice every 25 epochs, early stop after 5 checks without a 0.1 kcal/mol improvement; three checkpoints are kept (final, lowest training loss, lowest validation MAE), and the figures of Secs. 3 and 4 read the validation-best one. Over the 18 G1 cells the runs took 150 to 200 epochs (early-stopped in 12) and 1.2 to 22 h of wall each (`train_metadata.json`).

3.2 The differentiable SCF

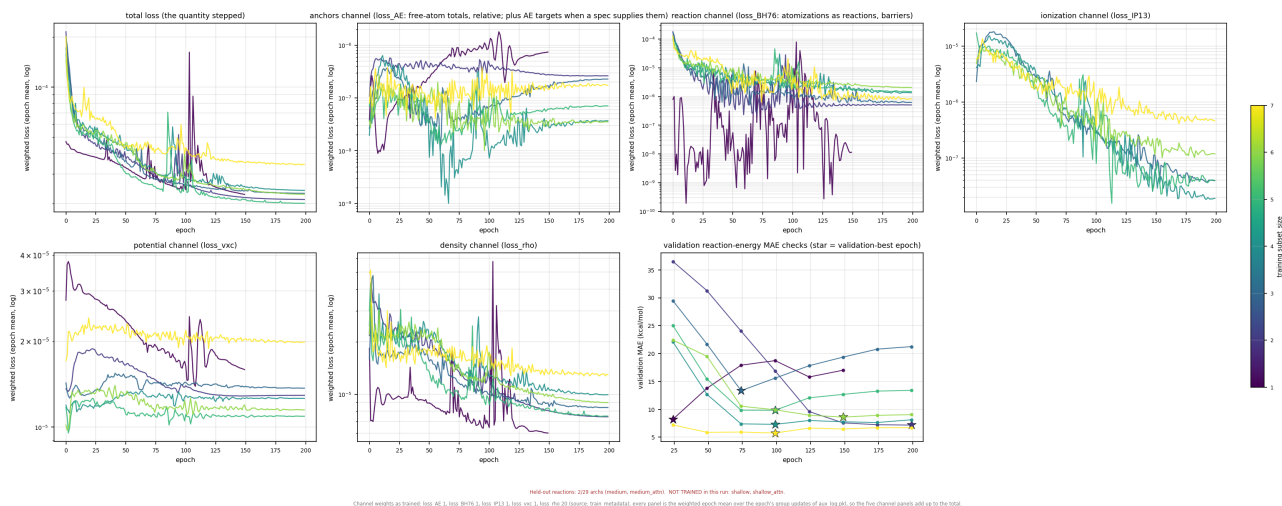
The energies and densities above are self-consistent: the manual backend runs a fixed $N = 3$ cycles from the converged PBE density (`solver:full_3`), diagonalizing the Kohn-Sham matrix built from the network's V_{xc} each cycle, with the DFS decaying linear mixing $D_{t+1} = (1 - a_t)D_t + a_t D_t^{\text{out}}$, $a_t = 0.3^t + 0.3$, a freeze once $|\Delta E| < 10^{-6}$ Ha, and gradients through every cycle. The density scored by $\mathcal{L}_\rho$ and by the density metrics is the last mixed density; three cycles do not converge the SCF for a functional far from a fixed point, which is the subject of the two arms submitted on 2026-09-08 (25 cycles; `dpyscf` parity) and of the converged-SCF re-evaluation in flight. The 2026-09-07 audit (`notebooks/analysis/AUDIT_2026-09-01.md`, the density section) records the full parity table against `dpyscf`.

3.3 Figures

Fig. 3.1 Per-channel training losses, one figure per architecture: the total and the five channels at their trained weights as per-epoch means (log y; the five panels add up to the total), one curve per subset size, the validation-MAE checks with the validation-best epoch starred, from each cell's `aux_log.pkl`.



Per-channel training losses -- medium_attn (weighted epoch means; panels sum to the total) · run_20260902T145245Z · val-best



Per-channel training losses -- deep_3x16 (weighted epoch means; panels sum to the total) · run_20260902T145247Z · val-best

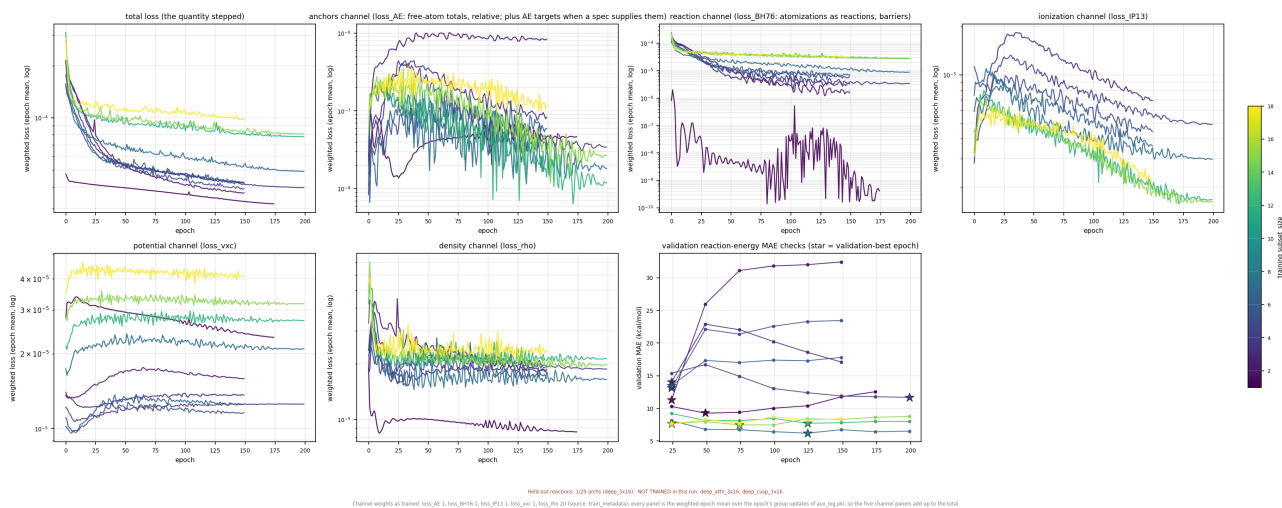
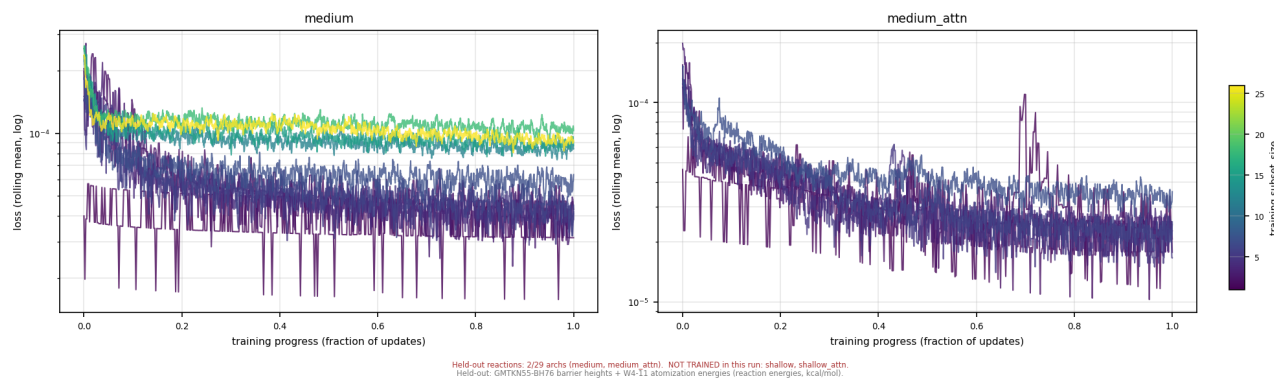


Fig. 3.2 Total training loss per cell (rolling mean, log-y), the loss actually stepped.

Training-loss trajectories by architecture (per-subset) · run_20260902T145245Z · val-best



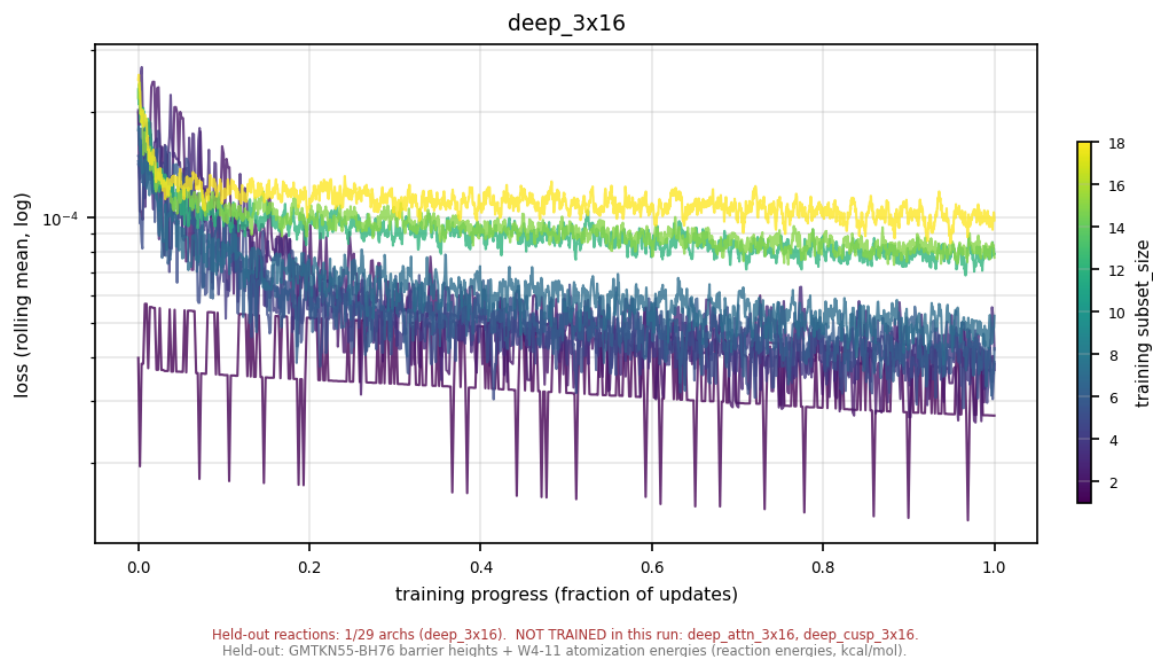
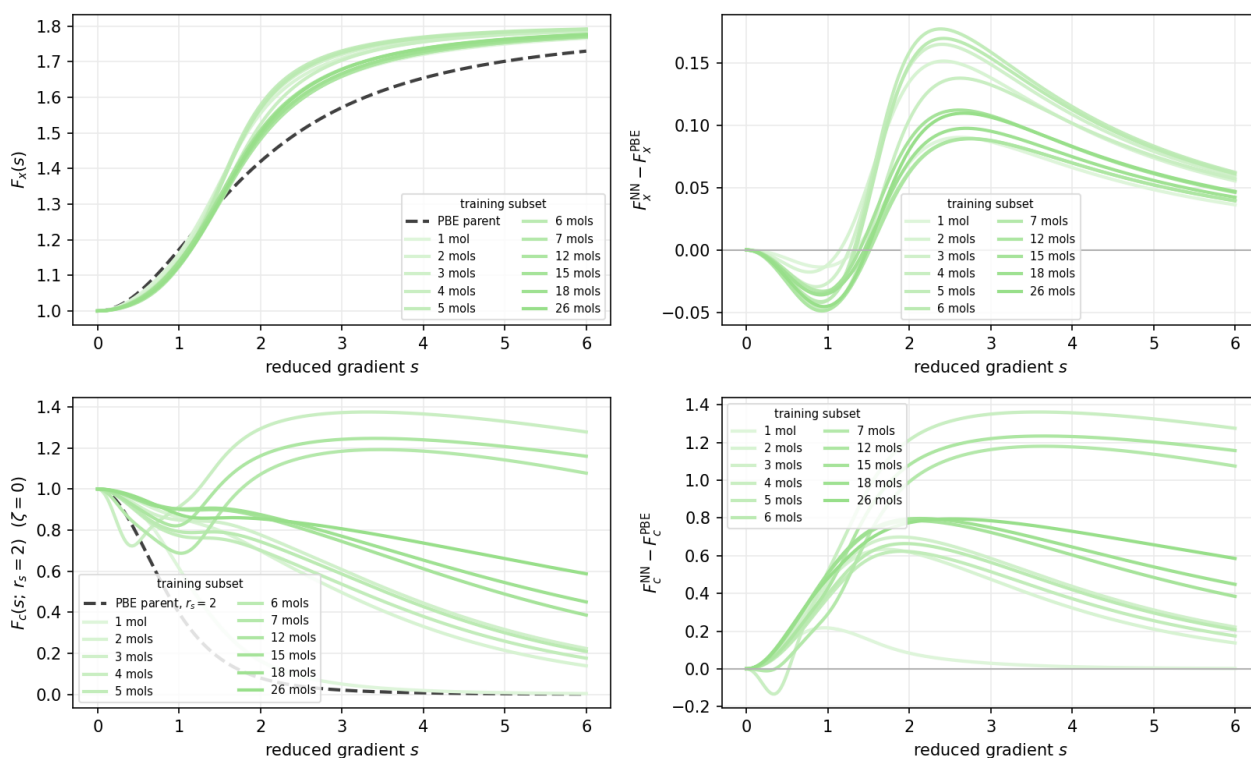


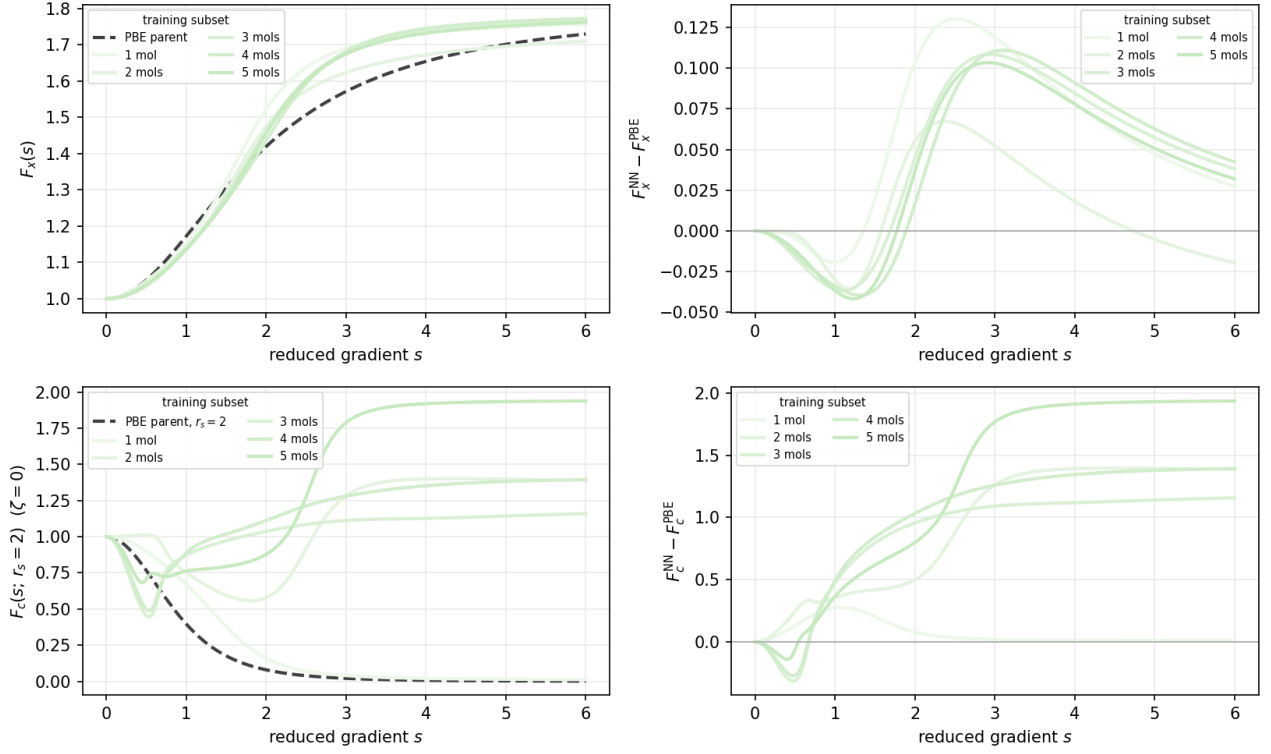
Fig. 3.3 The trained enhancement factors against PBE, one curve per subset size (validation-best checkpoint): $F_x(s)$ at $n = 1$ and $F_c(s; r_s = 2)$ at $\zeta = 0$, with the differences. The pre-trained curve of Fig. 2.3 is the starting point of every curve here.

medium: trained networks against the PBE parent



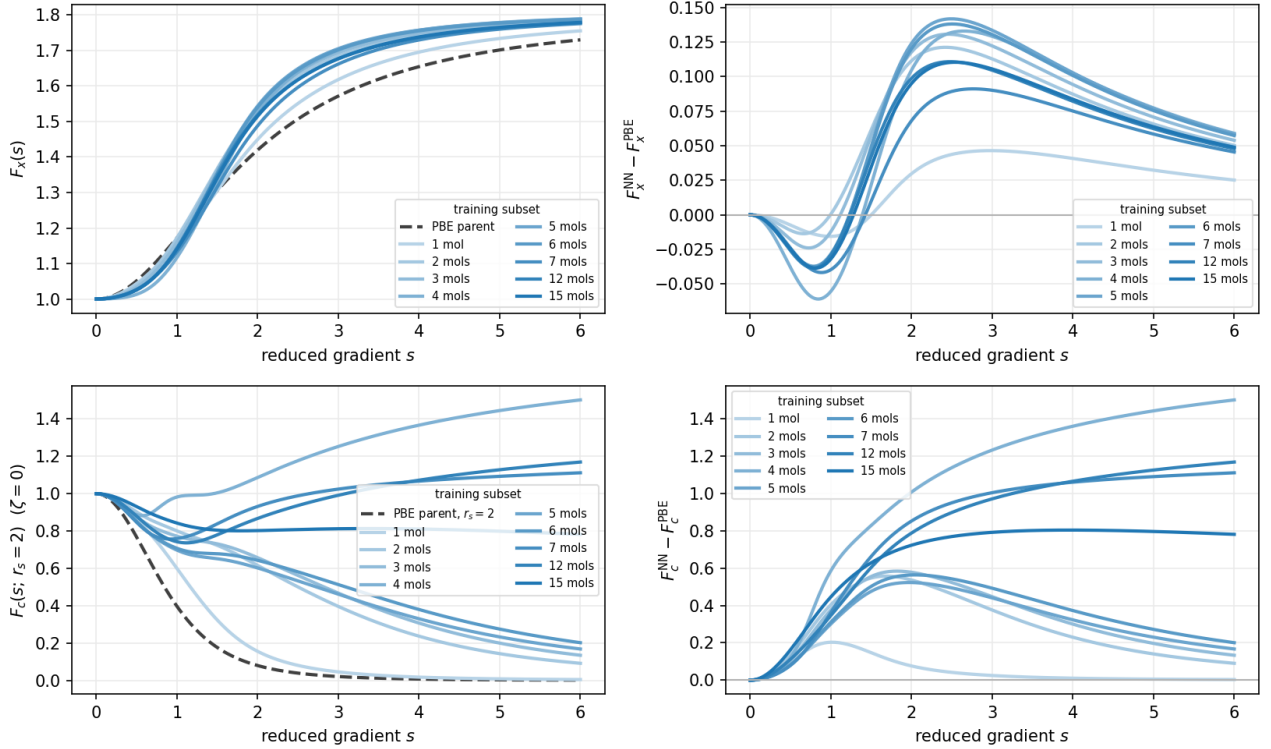
run_20260902T145245Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 11 completed cell(s); max|dF_x| 1.78e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).

medium_attn: trained networks against the PBE parent



run_run_20260902T145245Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 5 completed cell(s); $\max|dF_x|$ 1.30e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).

deep_3x16: trained networks against the PBE parent



run_run_20260902T145247Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 9 completed cell(s); $\max|dF_x|$ 1.42e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).

The training moves the exchange factor by up to 0.18 (medium, $s \approx 2.5$) and 0.14 (deep_3x16), a systematic raise between $s = 1.5$ and 6 with a dip below 1.5, similar in shape across subset sizes and largest at the small subsets; the correlation factor at $r_s = 2$ is raised by 0.5 to 1.4 for

18

Group	Architecture	Subset	Epochs	total first/last	AE first/last	reactions first/last	IP13 first/last	V_xc first/last	rho first/last	val MAE checks (kcal/mol)	val-best epoch
G1	medium	1	175	4.47e-05 / 3.66e-05	7.77e-08 / 1.93e-07	7.25e-07 / 4.16e-09	0.00e+00 / 0.00e+00	3.13e-05 / 2.68e-05	1.26e-05 / 9.62e-06	9.4, 8.2, 8.4, 9.1, 9.0, 9.1, 9.1	50
G1	medium	2	150	1.79e-04 / 4.29e-05	1.66e-07 / 7.16e-07	1.31e-04 / 4.49e-06	0.00e+00 / 0.00e+00	1.35e-05 / 1.73e-05	3.43e-05 / 2.05e-05	11.7, 25.2, 28.1, 30.0, 31.3, 32.7	25
G1	medium	3	150	1.83e-04 / 3.85e-05	4.91e-08 / 7.93e-08	1.36e-04 / 2.04e-06	5.12e-06 / 2.63e-06	1.39e-05 / 1.37e-05	2.82e-05 / 2.01e-05	14.6, 22.5, 22.0, 17.0, 16.0, 15.6	25
G1	medium	4	200	1.57e-04 / 4.19e-05	4.50e-08 / 3.98e-08	1.10e-04 / 3.05e-06	3.94e-06 / 2.06e-06	1.21e-05 / 1.33e-05	3.12e-05 / 2.35e-05	18.2, 20.0, 16.6, 15.6, 15.4, 14.9, 14.7, 14.6	200
G1	medium	5	150	1.39e-04 / 4.10e-05	3.75e-08 / 6.83e-08	9.16e-05 / 5.96e-06	1.08e-05 / 2.89e-06	1.07e-05 / 1.25e-05	2.62e-05 / 1.96e-05	19.2, 23.4, 23.3, 26.8, 27.6, 27.7	25
G1	medium	6	150	1.34e-04 / 4.51e-05	2.52e-08 / 7.90e-08	8.98e-05 / 8.13e-06	7.05e-06 / 2.11e-06	1.01e-05 / 1.33e-05	2.70e-05 / 2.15e-05	18.4, 21.1, 21.9, 22.6, 22.2, 22.4	25
G1	medium	7	200	1.58e-04 / 5.90e-05	1.29e-07 / 2.96e-08	9.01e-05 / 1.60e-05	8.12e-06 / 1.89e-06	1.69e-05 / 2.35e-05	4.29e-05 / 1.76e-05	7.8, 7.3, 7.4, 7.1, 6.9, 7.2, 6.9, 7.0	175
G1	medium	12	200	2.14e-04 / 8.38e-05	1.03e-07 / 4.38e-08	1.40e-04 / 3.07e-05	4.65e-06 / 1.56e-06	2.17e-05 / 2.79e-05	4.78e-05 / 2.36e-05	9.0, 8.5, 8.3, 8.7, 8.1, 7.8, 7.9, 7.9	150
G1	medium	15	150	1.96e-04 / 8.79e-05	1.03e-07 / 8.94e-08	1.15e-04 / 3.02e-05	4.00e-06 / 1.50e-06	2.92e-05 / 3.39e-05	4.74e-05 / 2.22e-05	7.3, 8.0, 7.6, 7.5, 7.9, 7.5	25
G1	medium	18	150	1.94e-04 / 1.03e-04	1.05e-07 / 2.08e-07	1.08e-04 / 3.13e-05	3.45e-06 / 3.78e-06	3.92e-05 / 4.31e-05	4.38e-05 / 2.46e-05	7.1, 8.0, 7.3, 8.0, 7.2, 7.3	25
G1	medium	26	175	1.79e-04 / 9.16e-05	1.98e-07 / 1.84e-07	9.34e-05 / 2.76e-05	3.08e-06 / 1.15e-06	3.73e-05 / 3.73e-05	4.54e-05 / 2.53e-05	9.3, 8.9, 9.3, 9.0, 10.4, 9.3, 9.7	50
G1	medium_attn	1	150	4.49e-05 / 2.28e-05	1.48e-07 / 7.30e-07	3.63e-07 / 1.39e-08	0.00e+00 / 0.00e+00	3.49e-05 / 1.60e-05	9.52e-06 / 6.03e-06	8.2, 13.8, 17.9, 18.7, 15.8, 17.0	25
G1	medium_attn	2	200	1.44e-04 / 2.11e-05	3.02e-07 / 2.63e-07	9.54e-05 / 5.08e-07	0.00e+00 / 0.00e+00	1.36e-05 / 1.29e-05	3.43e-05 / 7.44e-06	36.5, 31.3, 24.0, 16.9, 9.6, 7.6, 7.2, 7.2	200
G1	medium_attn	3	200	1.51e-04 / 2.30e-05	8.53e-08 / 2.28e-07	9.82e-05 / 6.22e-07	7.06e-06 / 3.99e-08	1.32e-05 / 1.37e-05	3.28e-05 / 8.41e-06	29.5, 21.7, 13.3, 15.6, 17.8, 19.4, 20.8, 21.2	75
G1	medium_attn	4	200	1.25e-04 / 2.40e-05	4.86e-08 / 3.67e-08	8.16e-05 / 1.36e-06	4.99e-06 / 1.87e-08	1.16e-05 / 1.26e-05	2.66e-05 / 9.94e-06	22.1, 12.7, 7.4, 7.3, 8.0, 7.8, 7.6, 8.1	100
G1	medium_attn	5	200	1.12e-04 / 2.00e-05	7.20e-08 / 7.07e-08	6.49e-05 / 1.49e-06	1.12e-05 / 4.01e-08	1.06e-05 / 1.09e-05	2.51e-05 / 7.51e-06	25.0, 15.4, 9.8, 9.8, 12.1, 12.7, 13.3, 13.4	100
G1	medium_attn	6	200	1.04e-04 / 2.27e-05	4.82e-08 / 3.52e-08	6.12e-05 / 2.06e-06	6.95e-06 / 1.19e-07	1.05e-05 / 1.15e-05	2.57e-05 / 8.98e-06	22.4, 19.5, 10.6, 9.9, 8.9, 8.6, 8.9, 9.0	150
G1	medium_attn	7	200	1.18e-04 / 3.43e-05	1.18e-07 / 1.77e-07	5.79e-05 / 8.12e-07	7.39e-06 / 4.76e-07	1.98e-05 / 1.98e-05	3.24e-05 / 1.30e-05	7.2, 5.9, 5.9, 5.7, 6.6, 6.5, 6.7, 6.7	100
G2a	deep_3x16	1	175	4.46e-05 / 3.20e-05	5.86e-08 / 4.65e-08	8.47e-07 / 3.81e-10	0.00e+00 / 0.00e+00	3.12e-05 / 2.33e-05	1.25e-05 / 8.61e-06	10.3, 9.3, 9.4, 10.0, 10.4, 11.8, 12.5	50
G2a	deep_3x16	2	150	1.80e-04 / 3.69e-05	1.37e-07 / 8.33e-07	1.35e-04 / 1.59e-06	0.00e+00 / 0.00e+00	1.33e-05 / 1.59e-05	3.09e-05 / 1.86e-05	11.2, 26.0, 31.1, 31.8, 32.0, 32.4	25
G2a	deep_3x16	3	150	1.82e-04 / 4.23e-05	3.82e-08 / 8.71e-08	1.37e-04 / 3.21e-06	4.91e-06 / 7.03e-06	1.35e-05 / 1.36e-05	2.60e-05 / 1.83e-05	14.0, 22.9, 22.0, 20.2, 18.6, 17.1	25
G2a	deep_3x16	4	200	1.55e-04 / 3.96e-05	4.02e-08 / 3.46e-08	1.10e-04 / 3.40e-06	3.92e-06 / 4.92e-06	1.18e-05 / 1.25e-05	2.93e-05 / 1.87e-05	15.4, 16.7, 14.9, 13.0, 12.4, 11.9, 11.8, 11.7	200
G2a	deep_3x16	5	150	1.38e-04 / 3.91e-05	3.06e-08 / 3.63e-08	9.35e-05 / 5.75e-06	1.00e-05 / 4.56e-06	1.04e-05 / 1.15e-05	2.42e-05 / 1.72e-05	13.1, 22.1, 21.4, 22.6, 23.3, 23.4	25
G2a	deep_3x16	6	150	1.33e-04 / 4.26e-05	2.23e-08 / 5.93e-08	9.04e-05 / 7.17e-06	6.70e-06 / 3.65e-06	9.91e-06 / 1.25e-05	2.57e-05 / 1.92e-05	13.5, 17.3, 17.0, 17.4, 17.3, 17.8	25
G2a	deep_3x16	7	200	1.53e-04 / 4.92e-05	1.42e-07 / 1.90e-08	8.72e-05 / 8.77e-06	8.61e-06 / 3.00e-06	1.71e-05 / 2.09e-05	4.03e-05 / 1.65e-05	8.1, 6.8, 6.7, 6.4, 6.2, 6.7, 6.4, 6.5	125
G2a	deep_3x16	12	200	2.11e-04 / 7.79e-05	1.22e-07 / 1.17e-08	1.36e-04 / 2.77e-05	4.97e-06 / 1.66e-06	2.21e-05 / 2.72e-05	4.71e-05 / 2.13e-05	9.2, 8.1, 8.1, 8.5, 7.7, 7.8, 8.0, 8.0	125
G2a	deep_3x16	15	200	1.93e-04 / 8.08e-05	1.30e-07 / 2.62e-08	1.13e-04 / 2.78e-05	4.52e-06 / 1.62e-06	2.93e-05 / 3.18e-05	4.60e-05 / 1.97e-05	7.6, 8.0, 7.5, 7.5, 8.4, 8.3, 8.6, 8.8	75
G2a	deep_3x16	18	150	1.86e-04 / 9.83e-05	1.42e-07 / 1.21e-07	1.01e-04 / 3.13e-05	4.22e-06 / 2.23e-06	3.84e-05 / 4.10e-05	4.26e-05 / 2.36e-05	7.6, 8.3, 7.7, 8.7, 8.2, 8.4	25

On the training reactions themselves (Fig. 3.5, the table below): every one of the 28 cells beats PBE on the DFS-units $\mathcal{ED}$ over its own training reactions and molecules; the training-reaction WTMAD-2 is 0.00 to 4.4 kcal/mol against 5.7 (G1) and 6.4 (G2a) for PBE on the union of the training reactions, the attention twin the tightest fit (0.05 to 1.77). The training-molecule ε_n sits above PBE’s pooled value in every medium cell but 1 and 26 and every deep_3x16 cell but 1 and 18, and below it in every medium_attn cell: the per-electron L1 and the grid RMSE of the previous paragraph rank the same fits differently, and the report keeps both.

In-sample per cell (insample_by_pool_3x3_eps.csv of each group’s _val_best set, identical in the final-step set; combined leg; E = 2-subset WTMAD-2 over the cell’s own training reactions, kcal/mol; eps over its training molecules; ED under gamma = 1084.87 kcal/mol; the PBE columns are the union of the run’s training reactions / molecules for E and eps and the cell’s own slice for ED; n = training reactions and trained species with a density)

Group	Architecture	Subset	n rxn	n species	E NN	E PBE (union)	eps NN	eps PBE (union)	ED NN	ED PBE (cell)	beats
G1	medium	1	1	1	0.02	5.67	0.00728	0.00888	0.05	0.79	True
G1	medium	2	2	3	2.14	5.67	0.01353	0.00888	3.73	11.27	True
G1	medium	3	2	3	2.96	5.67	0.01106	0.00888	4.75	12.57	True
G1	medium	4	3	4	2.63	5.67	0.01118	0.00888	4.33	11.58	True
G1	medium	5	3	4	4.31	5.67	0.01315	0.00888	6.63	13.25	True
G1	medium	6	4	5	3.92	5.67	0.01234	0.00888	6.06	11.78	True
G1	medium	7	5	6	2.04	5.67	0.01042	0.00888	3.46	7.89	True
G1	medium	12	10	11	3.11	5.67	0.01026	0.00888	4.86	8.86	True
G1	medium	15	13	14	2.78	5.67	0.00900	0.00888	4.32	8.08	True
G1	medium	18	16	16	2.60	5.67	0.00944	0.00888	4.14	8.01	True
G1	medium	26	24	22	2.23	5.67	0.00731	0.00888	3.49	7.14	True
G1	medium_attn	1	1	1	0.05	5.67	0.00494	0.00888	0.09	0.79	True
G1	medium_attn	2	2	3	0.59	5.67	0.00617	0.00888	1.08	11.27	True
G1	medium_attn	3	2	3	1.57	5.67	0.00626	0.00888	2.55	12.57	True
G1	medium_attn	4	3	4	1.68	5.67	0.00648	0.00888	2.71	11.58	True
G1	medium_attn	5	3	4	1.77	5.67	0.00732	0.00888	2.89	13.25	True
G1	medium_attn	6	4	5	1.72	5.67	0.00724	0.00888	2.83	11.78	True
G1	medium_attn	7	5	6	0.53	5.67	0.00640	0.00888	0.98	7.89	True
G2a	deep_3x16	1	1	1	0.00	6.44	0.00753	0.00888	0.00	0.79	True
G2a	deep_3x16	2	2	3	0.91	6.44	0.01392	0.00888	1.72	11.27	True
G2a	deep_3x16	3	2	3	3.81	6.44	0.01046	0.00888	5.71	12.57	True
G2a	deep_3x16	4	3	4	2.92	6.44	0.00969	0.00888	4.57	11.58	True
G2a	deep_3x16	5	3	4	4.36	6.44	0.01327	0.00888	6.69	13.25	True
G2a	deep_3x16	6	4	5	3.61	6.44	0.01257	0.00888	5.71	11.78	True
G2a	deep_3x16	7	5	6	1.58	6.44	0.01038	0.00888	2.77	7.89	True
G2a	deep_3x16	12	10	11	3.00	6.44	0.01037	0.00888	4.74	8.86	True
G2a	deep_3x16	15	13	14	2.72	6.44	0.00913	0.00888	4.27	8.08	True
G2a	deep_3x16	18	16	16	2.66	6.44	0.00873	0.00888	4.16	8.01	True

Group	Architecture	Subset	Trained species	Density RMSE NN	PBE	NN/PBE	Species improved	Loss fall (first/last 5%)	Updates
G1	medium	1	1	9.52e-05	1.32e-04	0.72	1/1	1.2	525
G1	medium	2	3	8.28e-05	1.03e-04	0.81	2/3	3.7	600
G1	medium	3	3	1.29e-04	1.33e-04	0.97	2/3	4.4	750
G1	medium	4	4	1.16e-04	1.22e-04	0.96	2/4	3.4	1200
G1	medium	5	4	9.34e-05	1.03e-04	0.90	2/4	3.5	1050
G1	medium	6	5	8.99e-05	1.02e-04	0.88	3/5	2.7	1200
G1	medium	7	6	1.03e-04	1.32e-04	0.78	5/6	2.2	1800
G1	medium	12	11	1.17e-04	1.49e-04	0.79	8/11	2.0	2800
G1	medium	15	14	1.10e-04	1.45e-04	0.76	11/14	1.9	2550
G1	medium	18	16	1.21e-04	1.52e-04	0.79	13/16	1.6	3000
G1	medium	26	22	1.18e-04	1.60e-04	0.73	20/22	1.6	4900
G1	medium_attn	1	1	7.52e-05	1.32e-04	0.57	1/1	1.9	450
G1	medium_attn	2	3	7.11e-05	1.03e-04	0.69	3/3	5.1	800
G1	medium_attn	3	3	8.95e-05	1.33e-04	0.67	3/3	4.8	1000
G1	medium_attn	4	4	8.30e-05	1.22e-04	0.68	4/4	4.1	1200
G1	medium_attn	5	4	6.81e-05	1.03e-04	0.66	4/4	4.3	1400
G1	medium_attn	6	5	6.61e-05	1.02e-04	0.65	5/5	3.7	1600
G1	medium_attn	7	6	1.00e-04	1.32e-04	0.76	6/6	2.9	1800
G2a	deep_3x16	1	1	9.00e-05	1.32e-04	0.68	1/1	1.4	525
G2a	deep_3x16	2	3	8.04e-05	1.03e-04	0.78	2/3	4.3	600
G2a	deep_3x16	3	3	1.23e-04	1.33e-04	0.93	2/3	4.0	750
G2a	deep_3x16	4	4	1.05e-04	1.22e-04	0.86	3/4	3.5	1200
G2a	deep_3x16	5	4	9.19e-05	1.03e-04	0.89	2/4	3.7	1050
G2a	deep_3x16	6	5	8.72e-05	1.02e-04	0.86	4/5	2.8	1200
G2a	deep_3x16	7	6	1.03e-04	1.32e-04	0.78	5/6	2.6	1800
G2a	deep_3x16	12	11	1.14e-04	1.49e-04	0.77	10/11	2.1	2800
G2a	deep_3x16	15	14	1.06e-04	1.45e-04	0.73	13/14	1.9	3400
G2a	deep_3x16	18	16	1.17e-04	1.52e-04	0.77	13/16	1.7	3000

The in-sample atomization-energy panels of the suite’s overview figure (insample_overview.png, panels A and B) are excluded from this report: that metric subtracts the network’s molecular energy from the fixed exact-atom anchors, so it reports the network’s absolute-energy offset (30 to 110 kcal/mol here) rather than an atomization error; the trained objective compares molecules with the network’s own atoms (Sec. 3.1). Fig. 3.5 replaces it with the residuals the training minimized.

4 Optimization, hold-out

Sources: xcquinox/alec/evaluation.py (the density metrics), notebooks/analysis/HOLDOUT_SET.md (the set), notebooks/analysis/make_ablation_arch_figure.py (the reductions, WTMAD-2 and ED), per cell eval_holdout_val_best/per_reaction.json and per_molecule.json, and the CSVs written beside the figures.

4.1 The held-out set and the comparators

The held-out pool is the union of the GMTKN55 BH76 barrier heights and the W4-11 atomization energies. Its test slice, after the removal of the 47-reaction validation slice (which selects the validation-best checkpoint and never enters a reported number), of each cell’s verbatim training reactions, and of the four barriers that appear twice under permuted-reactant names, carries 58 BH76 and 120 W4-11 reactions (name-deduplicated; the live count per cell is the `n_reactions` column of the tables below), and the density comparison runs over 189 molecular species with CCSD reference densities at the same basis and grid (atoms excluded). PBE is evaluated on exactly the same slice as each cell, so every comparison is like against like; the pooled PBE values are 7.98 (BH76), 13.74 (W4-11) and 11.86 kcal/mol (combined) reaction-energy MAE, and 8.92 kcal/mol 2-subset WTMAD-2. The functionals are evaluated with the same 3-cycle SCF they were trained with (Sec. 3.2); the PBE density is the converged reference calculation’s.

4.2 Metrics

Energy. Per-reaction absolute error $|E_{\text{rxn}}^{\text{NN}} - E_{\text{rxn}}^{\text{ref}}|$, reduced either as the plain MAE over a pool or as the two-bucket WTMAD-2 of GMTKN55 (Goerigk et al., PCCP 19, 32184 (2017)),

$$\text{WTMAD-2} = \frac{56.84}{N} \sum_{i \in \{\text{BH76}, \text{W4-11}\}} N_i \frac{\text{MAD}_i}{|\Delta E_{\text{ref}}|_i},$$

a labeled reweighting of the two present subsets, not the full-GMTKN55 quantity (the figures say so). **Density.** Per species, the grid-weighted RMSE $\sqrt{\sum_g w_g (n_g - n_g^{\text{ref}})^2} / \sqrt{\sum_g w_g}$ (the self-calibrated figures) and the DFS Eq. 20 per-electron error

$$\varepsilon_n = \frac{1}{N_e} \int |n - n^{\text{ref}}| d^3r, \quad N_e = \int n^{\text{ref}} d^3r$$

(the DFS-units figures), each reduced to a cell mean over species with case twins collapsed.

Combined. The DFS Eq. 21 harmonic combination

$$\mathcal{ED} = \frac{2}{1/E + 1/(\gamma D)},$$

with E the WTMAD-2 energy leg and D the density leg. In DFS units $\gamma = 1084.87$ kcal/mol, the Letter’s published slope, is dimensionally valid on ε_n and is used as is; in the RMSE units the figures self-calibrate $\gamma = E_{\text{PBE}}/D_{\text{PBE}}$ from the pooled PBE anchors, so that $\mathcal{ED}_{\text{PBE}} = E_{\text{PBE}}$ by construction and the NN against PBE ranking is independent of the density scale. A cell “beats PBE” when its $\mathcal{ED}$ is below the PBE $\mathcal{ED}$ of the same slice.

4.3 Figures, G1 (medium, medium_attn) and G2a (deep_3x16)

Every figure exists for both groups in figures_dfs_step7_dfs6311_grid3_v7g1_size_val_best/ and figures_dfs_step7_dfs6311_grid3_v7g2a_families_core_val_best/; the tail-excluded siblings are in the _excl_tail directories beside them.

Fig. 4.1 Held-out energy per cell: combined reaction-energy MAE (left) and 2-subset WTMAD-2 (right) against the pooled and cell-slice PBE values; the footer lists every training subset.

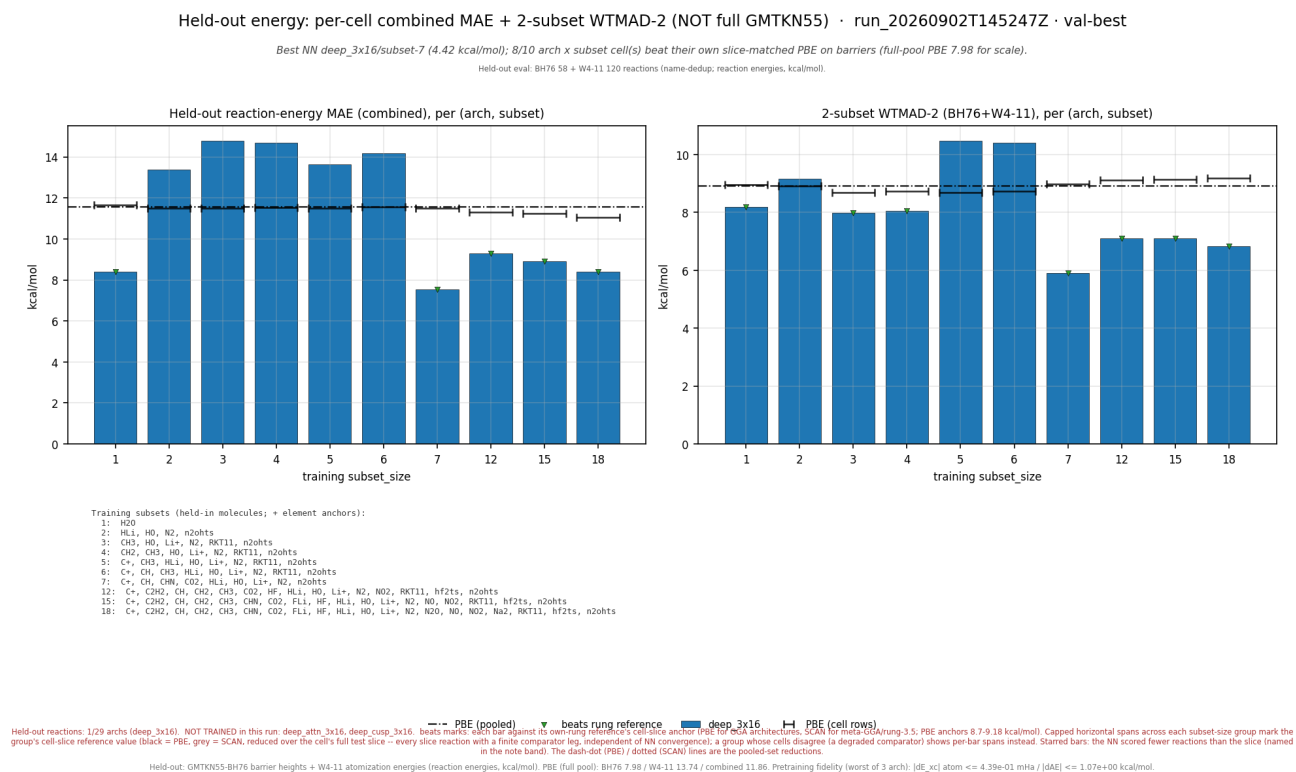
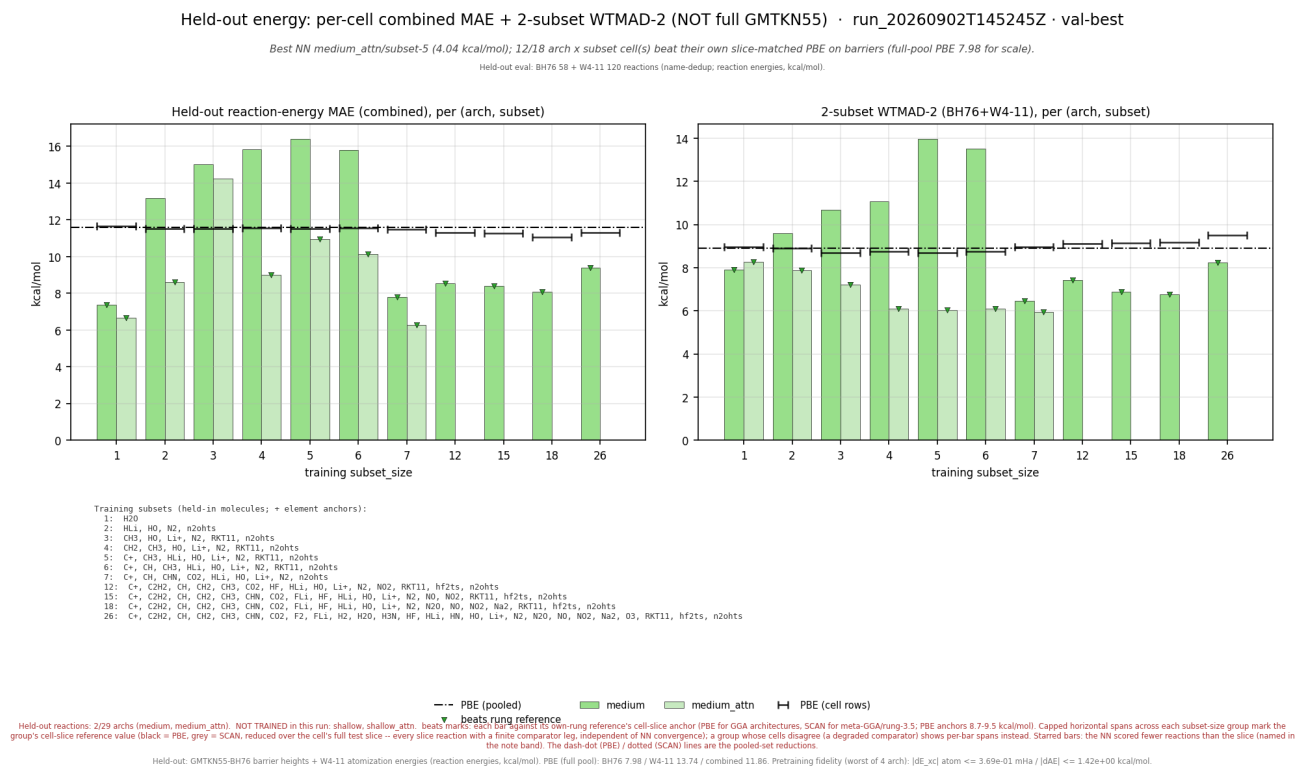


Fig. 4.2 The NN/PBE reaction-energy MAE ratio per cell (below 1 = the network is better).

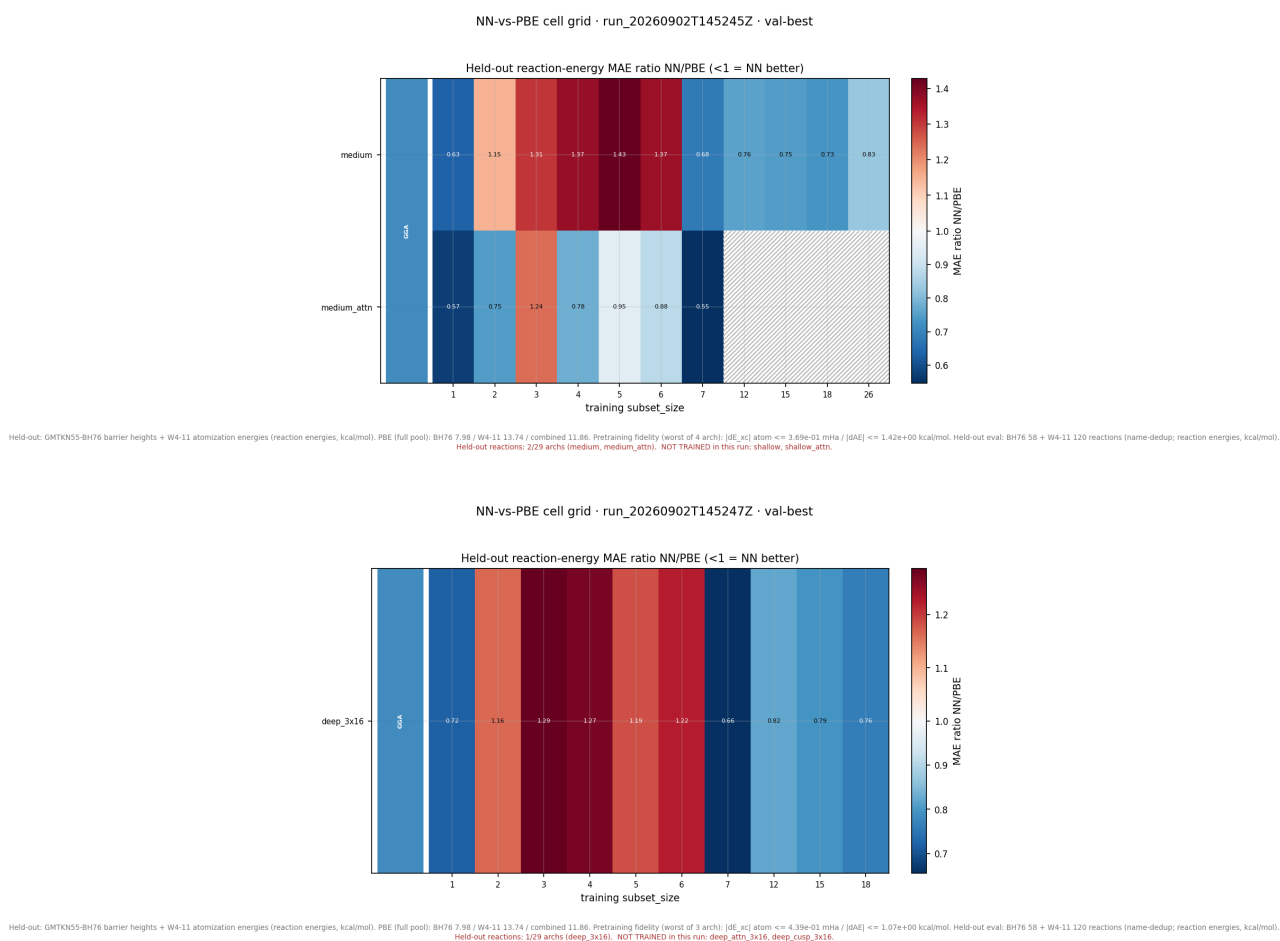
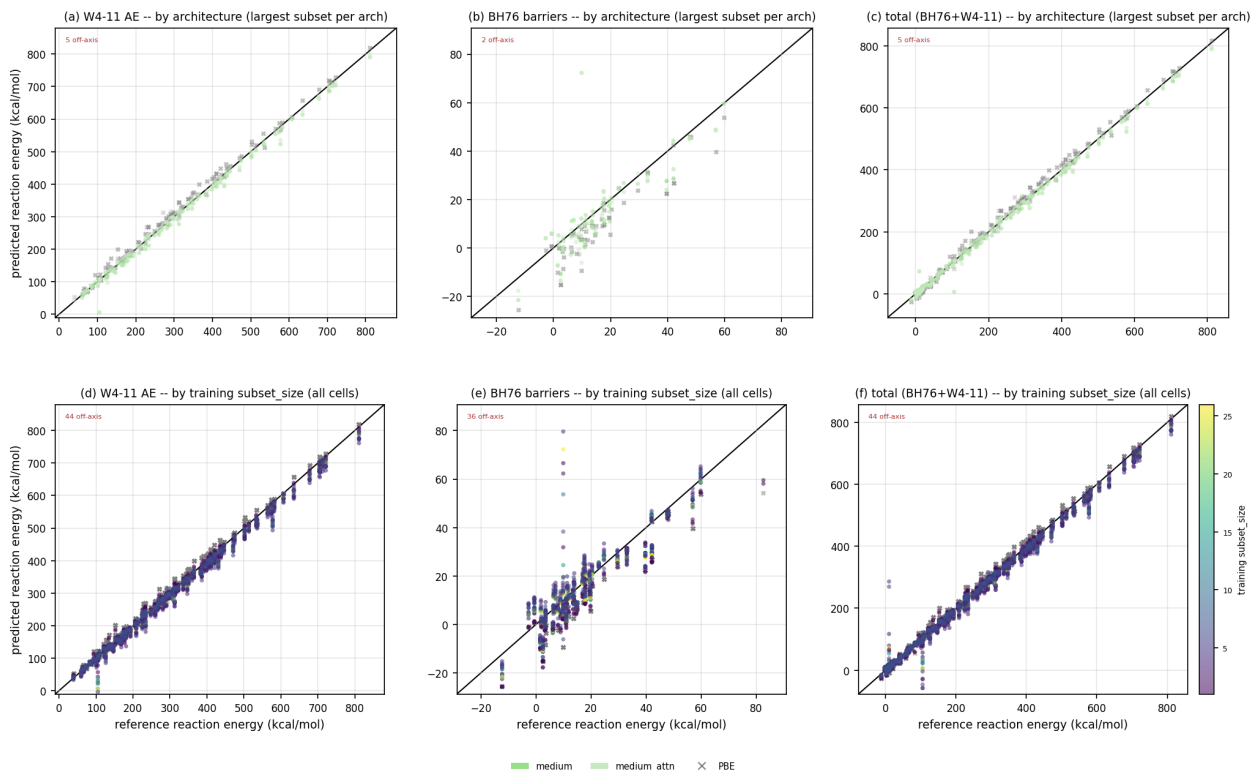


Fig. 4.3 Parity by reaction class (W4-11 atomizations, BH76 barriers, total), by architecture at its largest subset (top) and by subset size over all cells (bottom), PBE as grey crosses.

Held-out parity by reaction class -- AE | barriers | total · run_20260902T145245Z · val-best

Best NN medium_attn/subset-5 (4.04 kcal/mol); 12/18 arch x subset cell(s) beat their own slice-matched PBE on barriers (full-pool PBE 7.98 for scale).

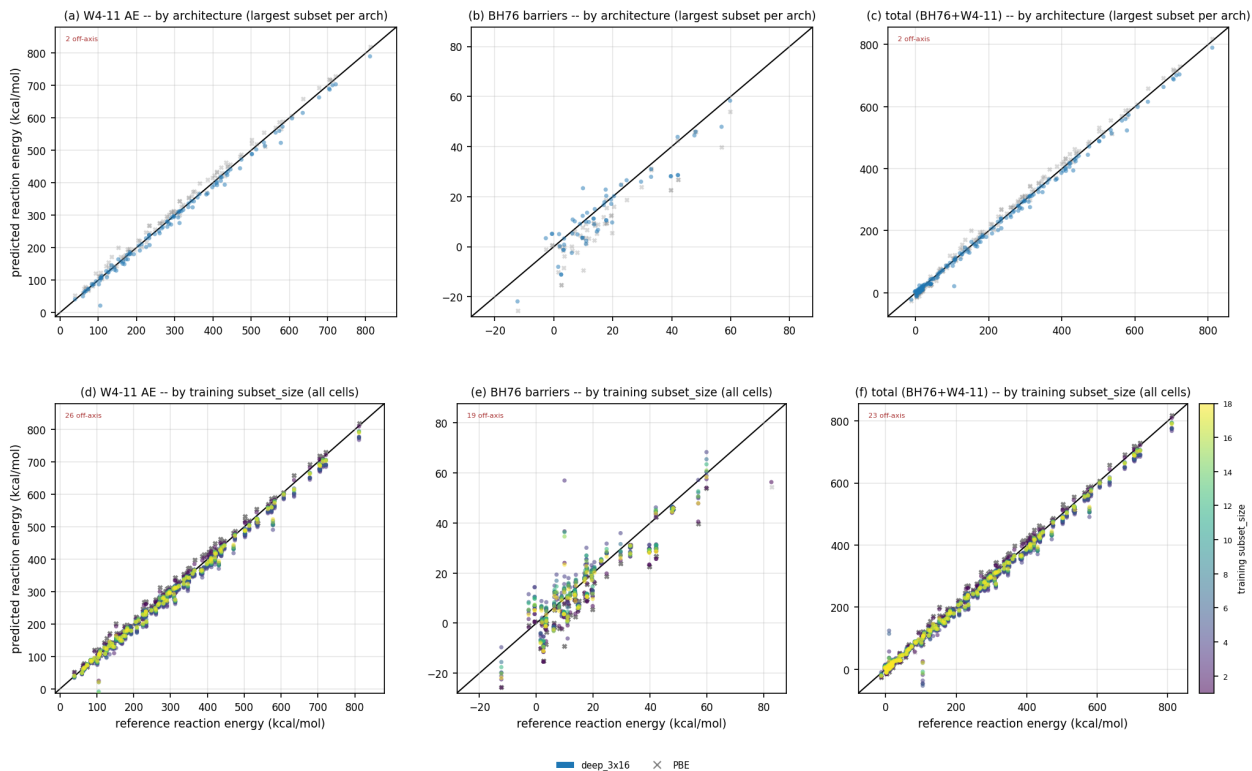


Held-out reactions: 2/29 archs (medium, medium_attn). NOT TRAINED in this run: shallow, shallow_attn.

Held-out: GMTKN55-BH76 barrier heights + W4-11 atomization energies (reaction energies, kcal/mol). PBE (full pool): BH76 7.98 / W4-11 13.74 / combined 11.86. Pretraining fidelity (worst of 4 arch): $|\Delta E_{xc}|_{atom} \leq 3.69e-01$ mHa / $|\Delta AE| \leq 1.42e+00$ kcal/mol. Held-out eval: BH76 58 + W4-11 120 reactions (name-depup; reaction energies, kcal/mol).

Held-out parity by reaction class -- AE | barriers | total · run_20260902T145247Z · val-best

Best NN deep_3x16/subset-7 (4.42 kcal/mol); 8/10 arch x subset cell(s) beat their own slice-matched PBE on barriers (full-pool PBE 7.98 for scale).



Held-out reactions: 1/29 archs (deep_3x16). NOT TRAINED in this run: deep_attn_3x16, deep_cusp_3x16.

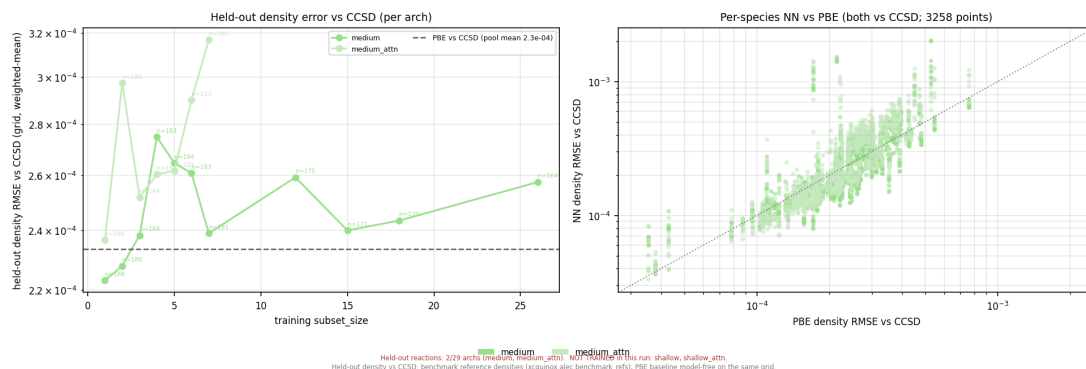
Held-out: GMTKN55-BH76 barrier heights + W4-11 atomization energies (reaction energies, kcal/mol). PBE (full pool): BH76 7.98 / W4-11 13.74 / combined 11.86. Pretraining fidelity (worst of 3 arch): $|\Delta E_{xc}|_{atom} \leq 4.39e-01$ mHa / $|\Delta AE| \leq 1.07e+00$ kcal/mol. Held-out eval: BH76 58 + W4-11 120 reactions (name-depup; reaction energies, kcal/mol).

Fig. 4.4 Held-out density against CCSD, network and PBE: the per-cell trend (left) and the per-species parity (right); the full set, then the same figure with the recurring tail removed (the

species above 1.5 x PBE in at least half of the cells that carry them and in at least two, listed in Sec. 4.5; a diagnostic view selected by the errors themselves, as its footer states).

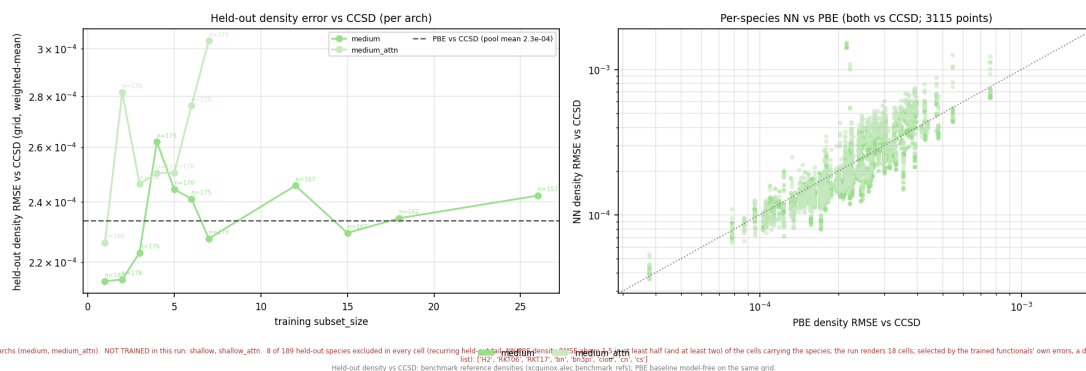
Held-out density error vs CCSD (NN vs PBE) · run_20260902T145245Z · val-best

HELD-OUT density error vs CCSD reference densities on the W4-11+BH76 benchmark species (atoms excluded; weighted-mean grid RMSE, NOT N_e-normalized). PBE baseline is model-free on the same grid. Density generalization; separate from the held-out ENERGY panels by design.
Held-out eval: BH76 58 + W4-11 120 reactions (name-dep; reaction energies, kcal/mol); density: 189 species vs CCSD refs at matching basis/grid (atoms excluded).



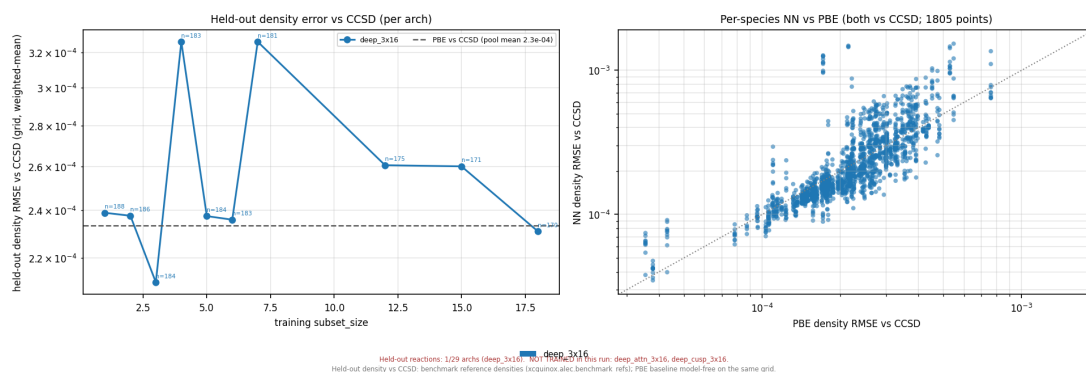
Held-out density error vs CCSD (NN vs PBE) · run_20260902T145245Z · val-best

HELD-OUT density error vs CCSD reference densities on the W4-11+BH76 benchmark species (atoms excluded; weighted-mean grid RMSE, NOT N_e-normalized). PBE baseline is model-free on the same grid. Density generalization; separate from the held-out ENERGY panels by design.
Held-out eval: BH76 58 + W4-11 120 reactions (name-dep; reaction energies, kcal/mol); density: 189 species vs CCSD refs at matching basis/grid (atoms excluded).



Held-out density error vs CCSD (NN vs PBE) · run_20260902T145247Z · val-best

HELD-OUT density error vs CCSD reference densities on the W4-11+BH76 benchmark species (atoms excluded; weighted-mean grid RMSE, NOT N_e-normalized). PBE baseline is model-free on the same grid. Density generalization; separate from the held-out ENERGY panels by design.
Held-out eval: BH76 58 + W4-11 120 reactions (name-dep; reaction energies, kcal/mol); density: 189 species vs CCSD refs at matching basis/grid (atoms excluded).



HELD-OUT density error vs CCSD reference densities on the W4-11+BH76 benchmark species (atoms excluded; weighted-mean grid RMSE, NOT N_e -normalized). PBE baseline is model-free on the same grid. Density generalization; separate from the held-out ENERGY panels by design.
Held-out eval: BH76 58 + W4-11 120 reactions (name-dedup; reaction energies, kcal/mol); density: 182 species vs CCSD refs at matching basis/grid (atoms excluded).



Per-channel held-out story (DFS units): WTMAD-2 | ε_{ref} | $\varepsilon_{\text{Dref}}$ (BH76, W4-11, combined) · run_20260902T145245Z · val-best

Columns are columns: BWtR | RWtI-1 | combined ABW and the GW ED energy days use the SINGLE_POOL_WTTR-2D, which collapses to 58 BPWR_positives(rwt_pos), the pool's mean abs deviation; rescaled by its mean abs reference energy; a scaled relative error, NOT a newweighting (only the combined column weights); NOT full OFFSETS.
 $(C_{RW} + 20)E + 26(\text{percent } C_{RW}) \times (\frac{1}{N} \sum_{i=1}^N |x_i - \mu|) = 20$ per percent). ONE gamma SHARED by all channels, value + source stamped in each row as one fractional bit when the calibration curve exceeds the latter's capacity (threshold 1984.77) (columns across clusters: C_{RW} , C_{RW} of FIVE in 2 FIVE, density row = call mean $\pm$ one-species parity in bold; density parity by pool_avg).

Per-channel held-out story (DFS units): WTMAD-2 | $z_{t|t}$ | $\mathcal{ED}_{t|t}$ (BH76, W4-11, combined) · run_20260902T145245Z · val-best

[illegible]

Fig. 4.6 The held-out overview on one canvas: WTMAD-2 by pool, density parity, the $(E, \gamma D)$ plane with iso- $\mathcal{ED}$ contours (points below the diagonal are density-limited, above it energy-limited) and the $\mathcal{ED}$ headline; self-calibrated γ .

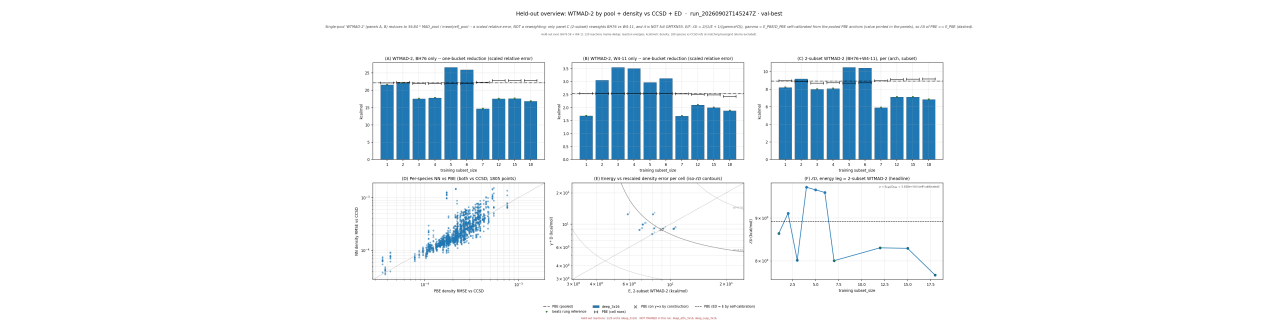


Fig. 4.7 The combined $\mathcal{ED}$ per cell with the self-calibrated γ , the decomposition plane, and the leg-independence check with the plain MAE as the energy leg.

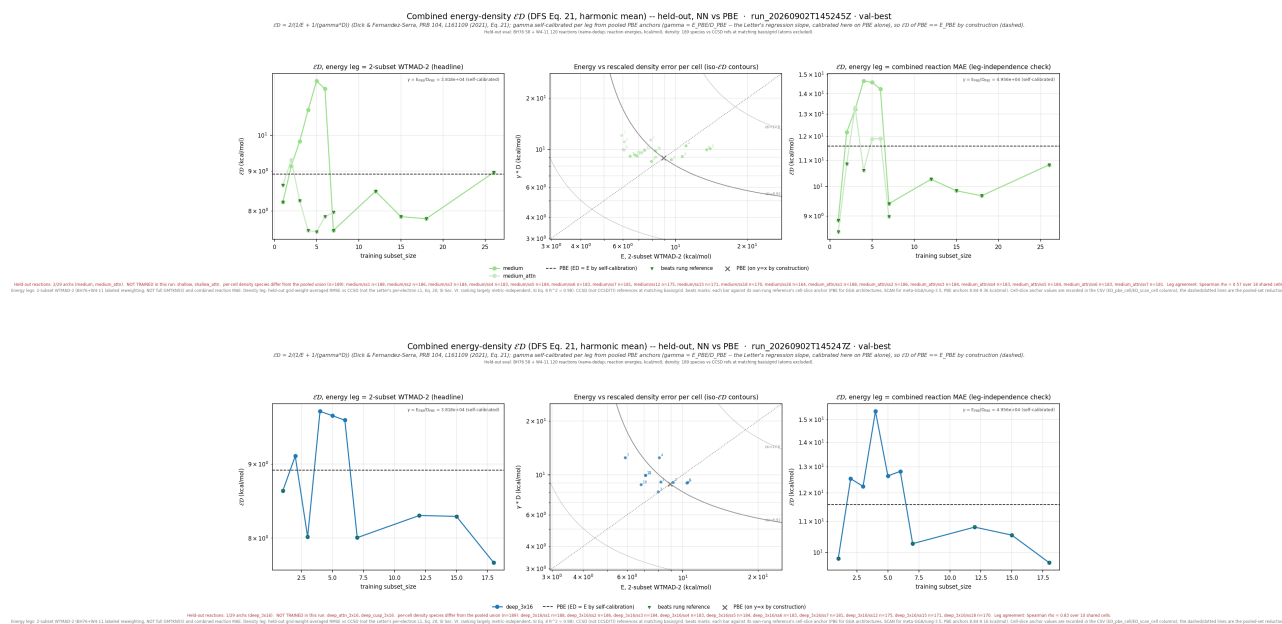
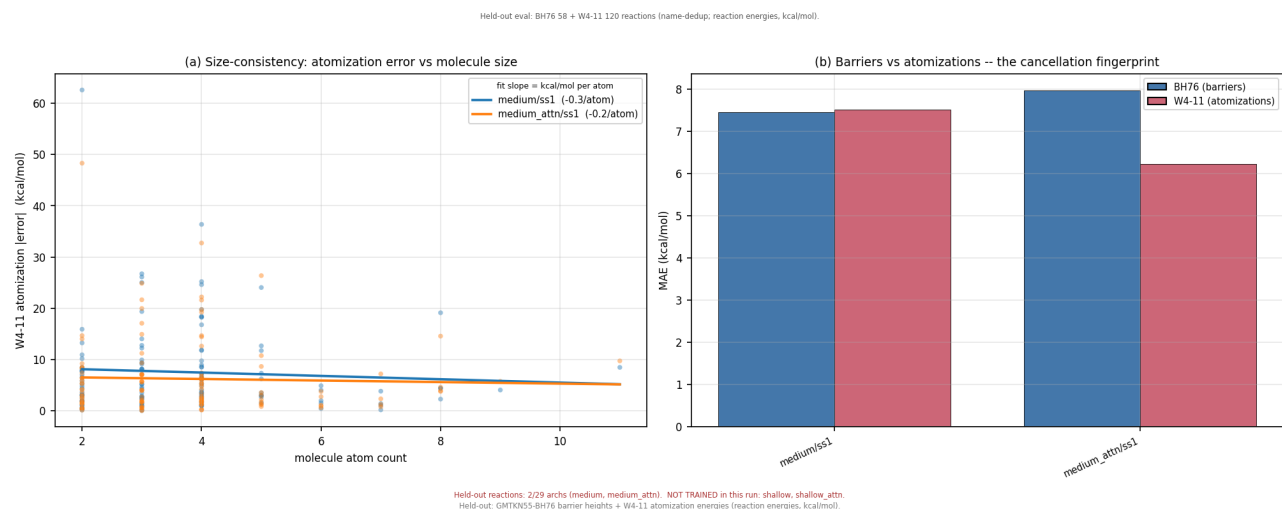


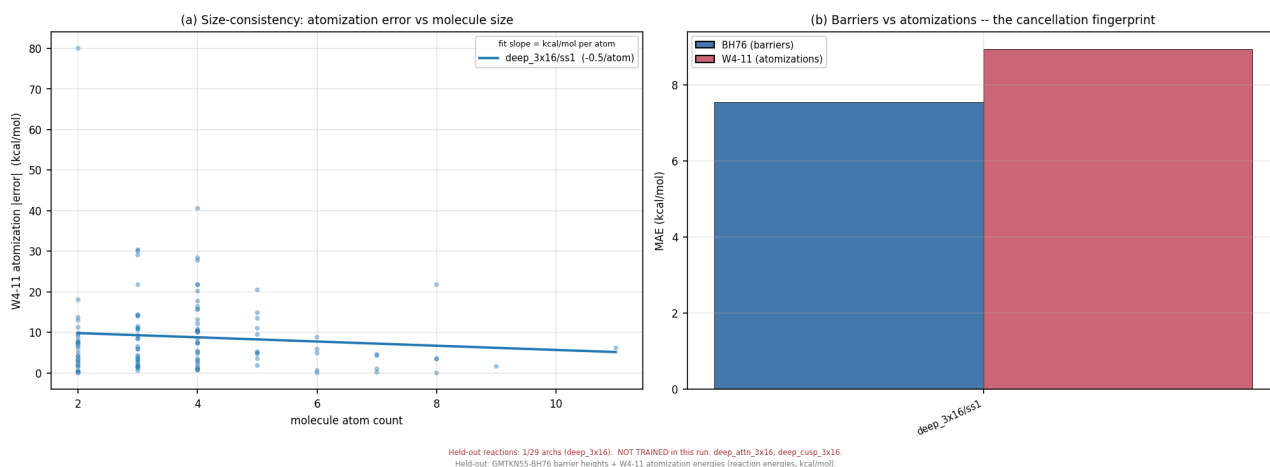
Fig. 4.8 Size-consistency diagnostic of the one-molecule cells: atomization error against molecule size, and barriers against atomizations.

Size-consistency (additivity) diagnostic -- medium/ss1, medium_attn/ss1 · run_20260902T1452452 · val-best



Size-consistency (additivity) diagnostic -- deep_3x16/ss1 · run_20260902T145247Z · val-best

Held-out eval: BH76 58 + W4-11 120 reactions (name-dedup; reaction energies, kcal/mol).



4.4 Results per cell

G1 (figures_dfs_step7_dfs6311_grid3_v7g1_size_val_best/holdout_by_pool_3x3_eps.csv and the _excl_tail sibling; combined leg; E = 2-subset WTMAD-2, kcal/mol; eps per electron; ED under gamma = 1084.87 kcal/mol; the PBE columns are the cell-slice values; "beats" = ED below the cell-slice PBE ED)

Architecture	Subset	n rxn	n species	E NN	E PBE	eps NN	eps PBE	ED NN	ED PBE	beats	eps NN (no tail)	eps PBE (no tail)	ED NN (no tail)	ED PBE (no tail)	beats (no tail)
medium	1	177	188	7.92	8.92	0.00951	0.00923	8.96	9.46	True	0.00886	0.00909	8.68	9.39	True
medium	2	177	186	9.61	8.92	0.01071	0.00923	10.52	9.44	False	0.00969	0.00909	10.04	9.37	False
medium	3	176	184	10.67	8.92	0.01068	0.00923	11.11	9.32	False	0.00959	0.00909	10.53	9.26	False
medium	4	175	183	11.06	8.92	0.01182	0.00923	11.88	9.34	False	0.01086	0.00909	11.41	9.28	False
medium	5	176	184	13.97	8.92	0.01113	0.00923	12.96	9.32	False	0.00981	0.00909	12.08	9.26	False
medium	6	175	183	13.51	8.92	0.01091	0.00923	12.62	9.34	False	0.00964	0.00909	11.79	9.28	False
medium	7	175	181	6.44	8.92	0.01017	0.00923	8.13	9.49	True	0.00933	0.00909	7.87	9.42	True
medium	12	169	175	7.41	8.92	0.01047	0.00923	8.97	9.56	True	0.00947	0.00909	8.61	9.49	True
medium	15	168	171	6.89	8.92	0.01004	0.00923	8.44	9.60	True	0.00917	0.00909	8.14	9.53	True
medium	18	166	170	6.74	8.92	0.00999	0.00923	8.31	9.62	True	0.00921	0.00909	8.05	9.55	True
medium	26	159	164	8.23	8.92	0.00973	0.00923	9.25	9.80	True	0.00876	0.00909	8.82	9.74	True
medium_attn	1	177	188	8.26	8.92	0.00925	0.00923	9.06	9.46	True	0.00858	0.00909	8.75	9.39	True
medium_attn	2	177	186	7.87	8.92	0.00972	0.00923	9.01	9.44	True	0.00846	0.00909	8.47	9.37	True
medium_attn	3	176	184	7.22	8.92	0.00981	0.00923	8.60	9.32	True	0.00936	0.00909	8.43	9.26	True
medium_attn	4	175	183	6.09	8.92	0.01020	0.00923	7.85	9.34	True	0.00929	0.00909	7.59	9.28	True
medium_attn	5	176	184	6.03	8.92	0.00883	0.00923	7.40	9.32	True	0.00785	0.00909	7.06	9.26	True
medium_attn	6	175	183	6.09	8.92	0.00973	0.00923	7.73	9.34	True	0.00859	0.00909	7.37	9.28	True
medium_attn	7	175	181	5.94	8.92	0.00963	0.00923	7.57	9.49	True	0.00852	0.00909	7.23	9.42	True

G2a (figures_dfs_step7_dfs6311_grid3_v7g2a_families_core_val_best/holdout_by_pool_3x3_eps.csv and the _excl_tail sibling; combined leg; E = 2-subset WTMAD-2, kcal/mol; eps per electron; ED under gamma = 1084.87 kcal/mol; the PBE columns are the cell-slice values; "beats" = ED below the cell-slice PBE ED)

Architecture	Subset	n rxn	n species	E NN	E PBE	eps NN	eps PBE	ED NN	ED PBE	beats	eps NN (no tail)	eps PBE (no tail)	ED NN (no tail)	ED PBE (no tail)	beats (no tail)
deep_3x16	1	177	188	8.18	8.92	0.00997	0.00923	9.32	9.46	True	0.00926	0.00908	9.02	9.39	True
deep_3x16	2	177	186	9.16	8.92	0.01143	0.00923	10.54	9.44	False	0.01040	0.00908	10.11	9.36	False
deep_3x16	3	176	184	7.99	8.92	0.01062	0.00923	9.44	9.32	False	0.00966	0.00908	9.07	9.25	True
deep_3x16	4	175	183	8.06	8.92	0.01160	0.00923	9.83	9.34	False	0.01064	0.00908	9.50	9.27	False
deep_3x16	5	176	184	10.48	8.92	0.01099	0.00923	11.16	9.32	False	0.00983	0.00908	10.57	9.25	False
deep_3x16	6	175	183	10.40	8.92	0.01097	0.00923	11.10	9.34	False	0.00984	0.00908	10.53	9.27	False
deep_3x16	7	175	181	5.90	8.92	0.01051	0.00923	7.77	9.49	True	0.00976	0.00908	7.58	9.41	True
deep_3x16	12	169	175	7.11	8.92	0.01058	0.00923	8.78	9.56	True	0.00957	0.00908	8.44	9.48	True
deep_3x16	15	168	171	7.10	8.92	0.01013	0.00923	8.63	9.60	True	0.00910	0.00908	8.26	9.52	True
deep_3x16	18	166	170	6.82	8.92	0.00996	0.00923	8.37	9.62	True	0.00913	0.00908	8.08	9.54	True

4.5 Reading

Energy. On the 2-subset WTMAD-2 the medium architecture beats PBE (8.92 kcal/mol) at 1 molecule and at every subset from 7 upward (6.44 to 8.23), and loses at 2 to 6 molecules (9.61 to 13.97); on the plain reaction-energy MAE (Fig. 4.2) the same cells read 0.63 and 0.68 to 0.83 of PBE against 1.15 to 1.43. The attention twin beats PBE at every one of its seven cells on WTMAD-2 (5.94 to 8.26), and on the MAE ratio at six of them (3 molecules: 1.24). `deep_3x16` is below PBE on WTMAD-2 at 1, 3, 4 and 7 to 18 (5.90 to 8.18) and above it at 2, 5 and 6 (9.16, 10.48, 10.40). The 2-to-6-molecule dip is common to the three architectures and to both groups; those subsets (HLi, HO, N₂, n2ohts at 2; CH₃, Li⁺, RKT11 added at 3; CH₂ at 4; C⁺ at 5; CH at 6, Fig. 4.1 footer) are the radical- and barrier-heavy ones, a plausible cause that this report does not test. From 7 molecules the correction generalizes, and the largest subsets do not improve on 7 to 18 (medium at 26: 8.23).

Density. In DFS units the held-out ε_n of the networks sits at or above PBE's (0.00923 per electron on the full slice) in every cell but one (medium_attn at 5 molecules, 0.00883), by 0.2 to 28 percent; the per-species parity (Figs. 4.4 and 4.6) shows the bulk of species near the diagonal and a tail well above it. The tail is recurring: eight species for G1 (bn, above 1.5 x PBE in all 18 cells with a median ratio of 6.3; rkt17 2.0; cloo 2.0; bn3pi 2.0; cn 1.9; h2 1.7; rkt06 1.6; cs 1.5) and seven for G2a (bn 6.6, rkt17 2.0, bn3pi 1.9, h2 1.8, cn 1.8, rkt06 1.8, cloo 1.8), computed from `eval_holdout_val_best/per_molecule.json` by the rule of Fig. 4.4. Without it the networks' ε_n moves to 0.0079 to 0.0109 against PBE's 0.0091 on the same reduced set, below PBE in medium_attn at 1, 2, 5, 6 and 7 and in medium at 1 and 26, within 1.5 percent of it in medium at 15 and 18 and in deep_3x16 at 15 and 18, and above it elsewhere; in the medium column from 7 molecules upward the ratio is 0.96 to 1.04. The tail species are the multireference and open-shell cases (BN and its 3-Pi state, CN, CS, ClOO, the OH + Cl and the H + CH₄ transition states, and H₂) where three SCF cycles from the PBE seed do not settle. The audit of 2026-09-07 found the comparison asymmetric by protocol (an unconverged network density against the converged PBE one) and could not decide from the pulled data whether the tail belongs to the functionals or to the three-cycle evaluation; the density channel itself trains (Sec. 3.4), and the converged-SCF re-evaluation in flight is the direct test.

Combined. On $\mathcal{ED}$ in DFS units, against the cell-slice PBE value: medium beats PBE at 1, 7, 12, 15, 18 and 26 and fails at 2 to 6; medium_attn beats it at every cell; deep_3x16 beats it at 1, 7, 12, 15 and 18 and fails at 2 to 6, where at 3 and 4 molecules the small energy margin (7.99 and 8.06 against 8.92) is outweighed by the density leg (0.0106 and 0.0116 against 0.0092). The verdicts are unchanged by removing the tail except deep_3x16 at 3 molecules, which then beats PBE (9.07 against 9.25). The wins are energy wins: on the decomposition plane every winning cell lies to the left of the PBE point, and its density leg is below PBE's only in medium_attn at 5 molecules. The best cells are medium_attn at 5 to 7 molecules ($\mathcal{ED}$ 7.40 to 7.73 against 9.32 to 9.49) and deep_3x16 at 7 (7.77 against 9.49).

5 Pending work and open items

- Cells: medium_attn at 12 to 26, deep_attn_3x16 and deep_cusp_3x16 at every subset, the shallow pair at every subset, are queued on the cluster; the figures and tables are regenerated from the same suite as they land.
- Meta-GGA group: four certificates pending; the deep_cusp_mgga_3x16 FAIL holds the training array (Sec. 2.4). The decision between dropping the architecture, refitting with more capacity or a higher energy-term weight, and the anchored class follows the four results.
- Converged-SCF re-evaluation of the 28 finished cells (the converged channel, DIIS at 10^{-8} Ha) and the CCSD T1 diagnostics for the model-free multireference list: two cluster jobs pending; they produce the `_excl_t1` and `_excl_unconverged` siblings of every density figure, the model-free counterparts of the `_excl_tail` view used here.
- The two training arms of the density program on the medium architecture at 7, 12, 15, 18 and 26 molecules: 25 SCF cycles with the current protocol (arm A) and the dpyscf-parity protocol (arm B: no convergence freeze, the executed first-mix index, the randomized

atomic/PBE seed mixture, Adam with the plateau schedule and the Letter's L2 and channel weights). Submitted 2026-09-08, no results yet.

- In-sample: the suite's in-sample atomization panels stay excluded until that metric uses the network's own atoms; Figs. 3.1 and 3.5 carry the trained quantities instead.